Cloud & Data Engineering - Introduction

IA Institut - 2025/2026

Aimene Belfodil

Section 0

Internship

Section 1

Real World Settings

A DAY IN DATA

The exponential growth of data is undisputed, but the numbers behind this explosion - fuelled by internet of things and the use of connected devcies – are hard to comprehend, particularly when looked at in the context of one day

DEMYSTIFIYING DATA UNITS

From the more familiar 'bit' or 'megabyte', larger units of measurement are more frequently being used to explain the masses of data

Unit		Value	Size
b	bit	0 or 1	1/8 of a byte
B	byte	8 bits	1 byte
KB	kilobyte	1,000 bytes	1,000 bytes
MB	megabyte	$1,000^2$ bytes	1,000,000 bytes
GB	gigabyte	$1,000^3$ bytes	1,000,000,000 bytes
TB	terabyte	$1,000^4$ bytes	1,000,000,000,000 bytes
PB	petabyte	$1,000^5$ bytes	1,000,000,000,000,000 bytes
EB	exabyte	$1,000^6$ bytes	1,000,000,000,000,000,000 bytes
ZB	zettabyte	$1,000^7$ bytes	1,000,000,000,000,000,000,000 bytes
YB	yottabyte	$1,000^8$ bytes	1,000,000,000,000,000,000,000,000 bytes

*A lowercase "b" is used as an abbreviation for bits, while an uppercase "B" represents bytes.

463EB

of data will be created every day by 2025

IDC

500m

tweets are sent every day

Twitter

4PB

of data created by Facebook, including

350m photos

100m hours of video watch time

Facebook Research

95m

photos and videos are shared on Instagram

Instagram Business

320bn

emails to be sent each day by 2021

294bn

billion emails are sent

Radicati Group

306bn

emails to be sent each day by 2020

65bn

messages sent over WhatsApp and two billion minutes of voice and video calls made

Facebook

3.9bn

people use emails

4TB

of data produced by a connected car

Intel

Searches made a day → **5bn**

Searches made a day from Google → **3.5bn**

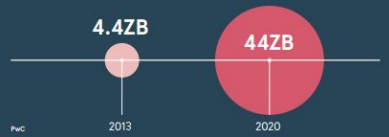
Smart insights

28PB

to be generated from wearable devices by 2020

Statista

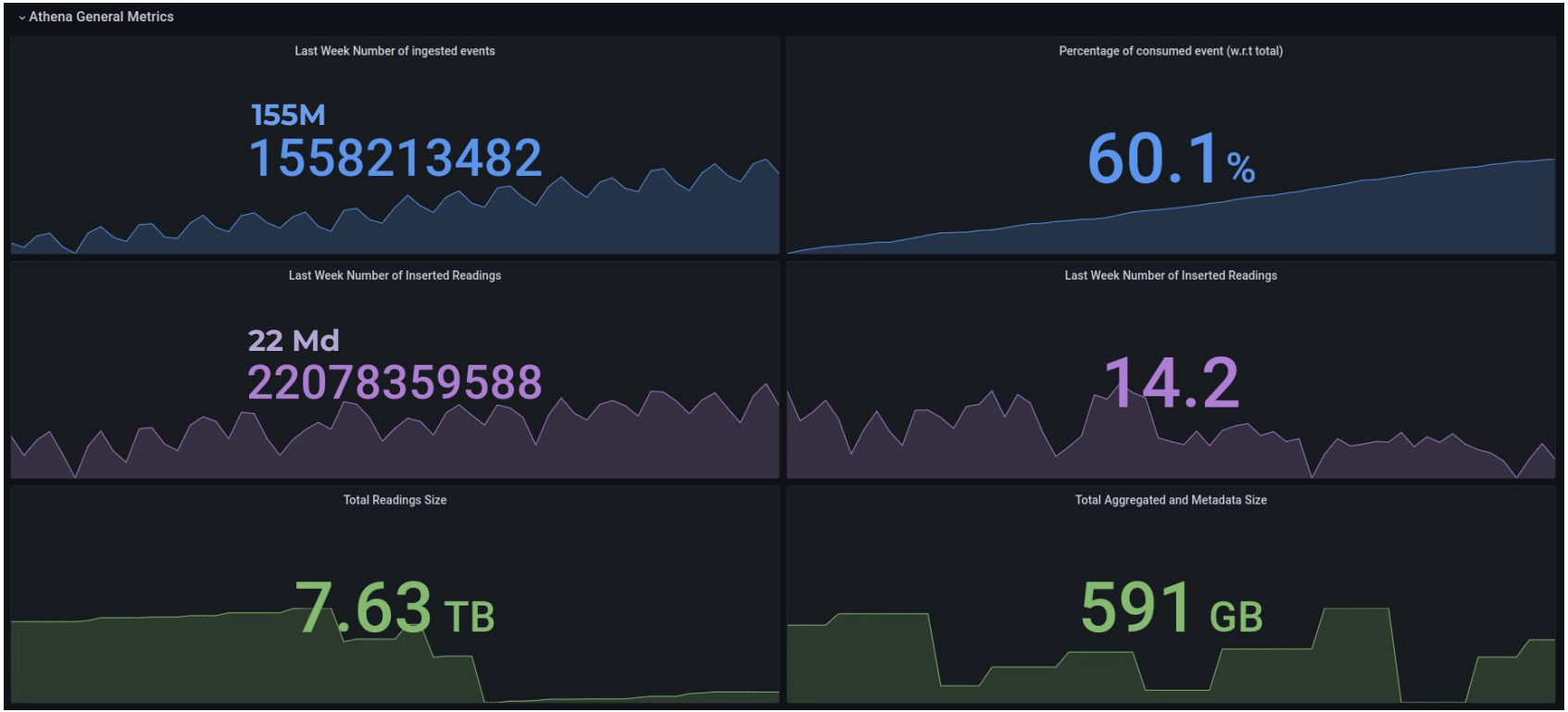
ACCUMULATED DIGITAL UNIVERSE OF DATA



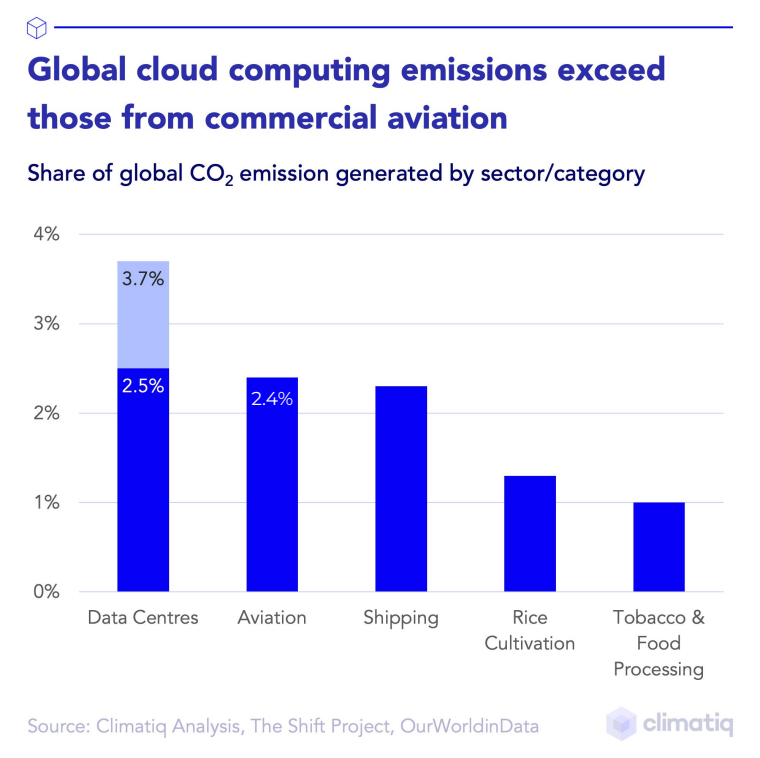
PwC

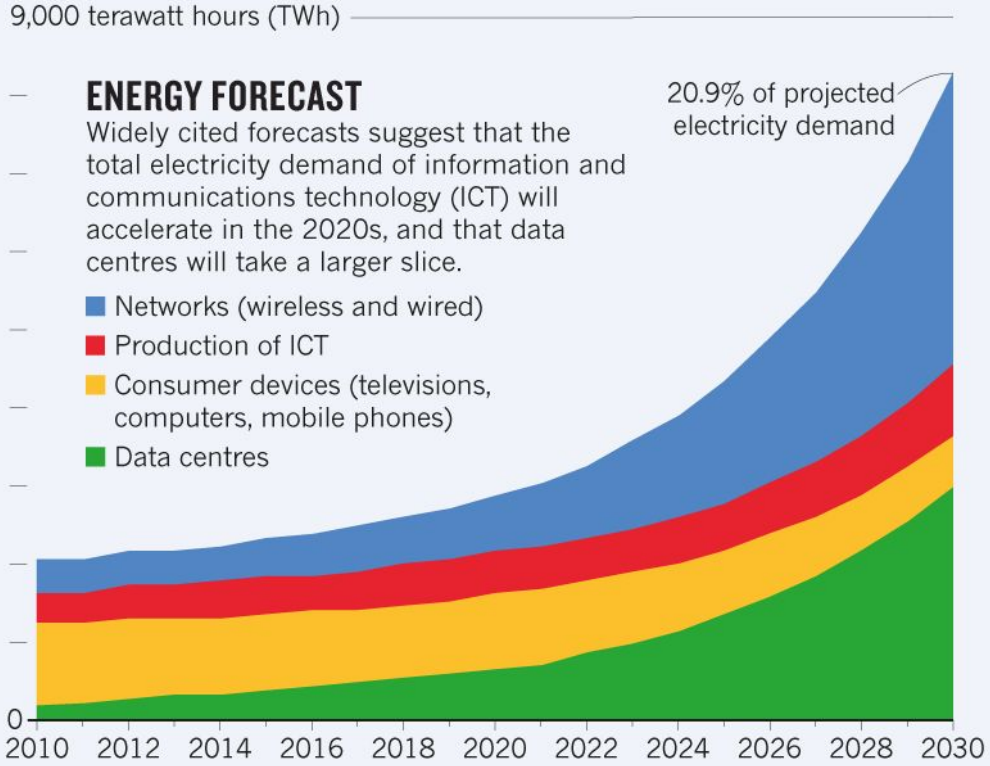
Source: https://www.raconteur.net/infographics/a-day-in-data/ RACONTEUR

Munic Use Case



Source: https://www.datacentermap.com/datacenters/

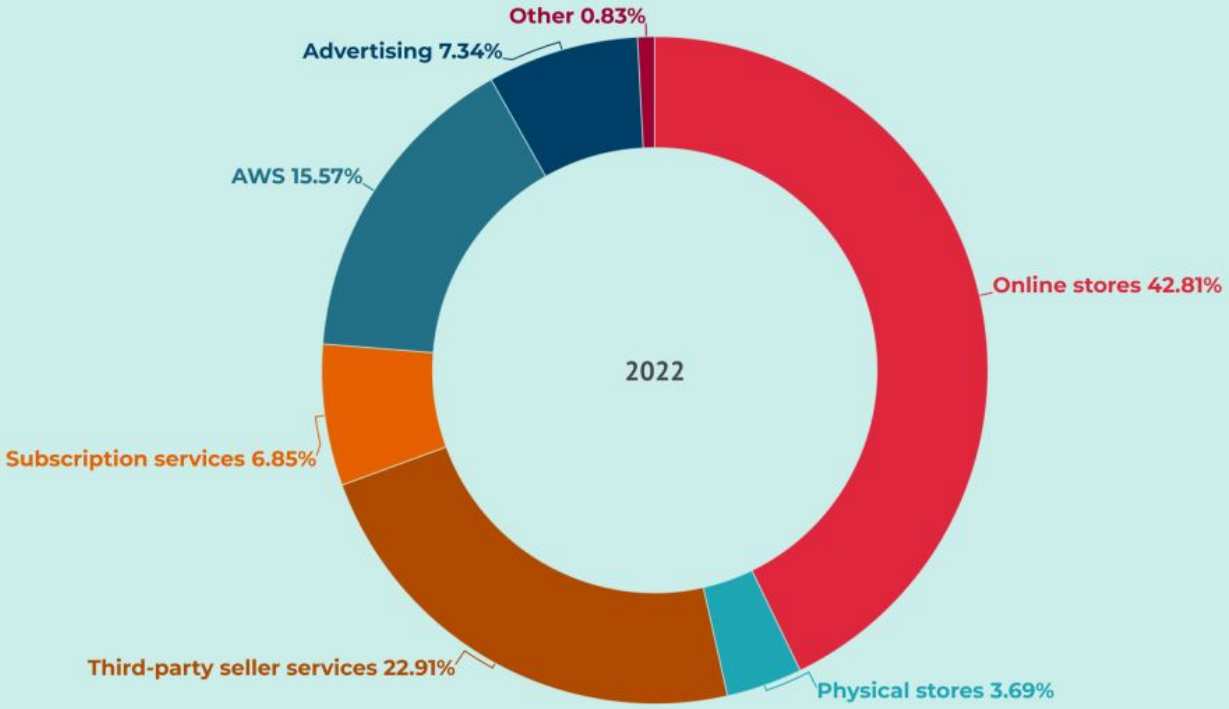




https://www.climatiq.io/blog/measure-greenhouse-gas-emissions-carbon-data-centres-cloud-computing (2022)
* https://**theshiftproject**.org/en/article/lean-ict-our-new-report/

https://www.akcp.com/blog/the-real-amount-of-energy-a-data-center-use/

Amazon Revenue Breakdown



80

Billions $

Cloud Computing as an important source of revenue

Source:
https://web.developpez.com/actu/318808/Un-incendie-pourrait-couter-105-millions-d-euros-a-OVHcloud-selon-les-documents-relatifs-a-l-introduction-en-bourse/

Source:
https://www.solutions-numeriques.com/exclusif-les-causes-de-lincendie-ovh-strasbourg-des-explications-stupefiantes/

3,6 millions de sites internet affectés

Section 2

Words that should be understood after this Curriculum

Notions à Apprendre

IAAS

Architecture

Backend

Broker

Scaleway

minikube

Docker

Virtualisation

Kubernetes

SAAS

PAAS

Kafka

AWS

API

OVH

Frontend

Cloud

GCP

Docker
Compose

Vagrant

Conteneurisation

Azure

Section 3

Cloud Computing

What do you understand by the term “Cloud” ?

Who coined the term “Cloud Computing” ?

What's interesting [now] is that there is an **emergent new model**, and you all are here because you are part of that new model. I don't think people have really understood how big this opportunity really is. It starts with the premise that the **data services** and **architecture** should be on servers. We call it **cloud computing** – they should be in a "**cloud**" somewhere. And that if you have the right kind of browser or the right kind of access, it doesn't matter whether you have a PC or a Mac or a mobile phone or a BlackBerry or what have you – or new devices still to be developed – you can get access to the cloud. There are a number of companies that have benefited from that. …

Eric Schmidt (CEO of Google) - 2006

Google Press Center

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August 9, 2006

Search Engine Strategies Conference
Conversation with Eric Schmidt hosted by Danny Sullivan
Eric Schmidt

Danny: I'd like to welcome all of you out here today. We're doing a little bit of a late start, but I figure after yesterday's Google Dance you could probably use a little bit of extra time. We are joined here today by Eric Schmidt, Chairman and CEO of Google. Thank you very much for coming out and being with us here today, Eric, we appreciate it.

Eric: Thank you very much.

Danny: I'd like to say hello to everybody who is listening to us live on the Internet. We're going out through Webmaster Radio FM, so if you're tuning in out there, hello to all of you as well. And now we're going to dive in with in the questions.

So in preparing for this, I was talking with a bunch of people at Google about things that would be interesting to address, things I was interested in, things that you're looking at doing. And they talked to me about how you're developing this idea of trying to help people not "bet against the Internet." And I almost threw up my hands and said, who's betting against the Internet? Does somebody think we're going to shove the genie in the bottle or change it? So tell us more about this.

Eric: Well, in the first place, thank you for having me. Congratulations on an amazing conference. This is really very much the center of the Web and of Web advertising today. I really appreciate you all coming and spending a couple of days, and I hope you enjoyed the party. And we certainly did as well.

It may be helpful to put architecture and the Internet in something of a historic context to start with, and say that 20 years ago everybody was in love with a concept called client/server computing. Many of the companies that are here in the [Silicon] Valley and in the industry as a whole were built up around a model where you had a PC client and then a set of servers, Unix, OS2, Windows, etc., and a lot of proprietary protocols between those. And many of those protocols and architectures are still present today.

What's even more interesting (and technical people never really think about this) is that there was also a business model built to fund and hire all those engineers to build all that. And the business model was largely invented by Oracle. It was a direct sales force that would go in and sell complicated software to enterprises that they would integrate and do important business functions.

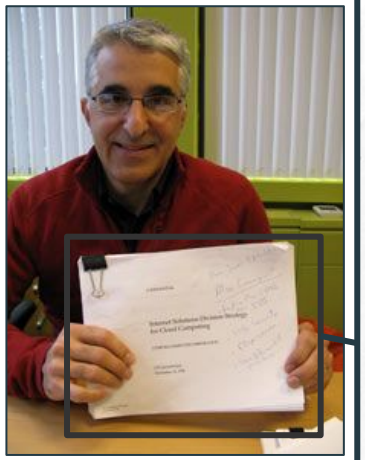
What's interesting [now] is that there is an emergent new model, and you all are here because you are part of that new model. I don't think people have really understood how big this opportunity really is. It starts with the premise that the data services and architecture should be on servers. We call it cloud computing – they should be in a "cloud" somewhere. And that if you have the right kind of browser or the right kind of access, it doesn't matter whether you have a PC or a Mac or a mobile phone or a BlackBerry or what have you – or new devices still to be developed – you can get access to the cloud. There are a number of companies that have benefited from that. Obviously, Google, Yahoo!, eBay, Amazon come to mind. The computation and the data and so forth are in the servers.

This is the same talk that I gave in this room 10 years ago about something they called the network computer – which, I can assure you, none of you are using, because it didn't work. But in the last 10 years, technology and architecture, things called Ajax and something called LAMP, which is Linux, Apache,

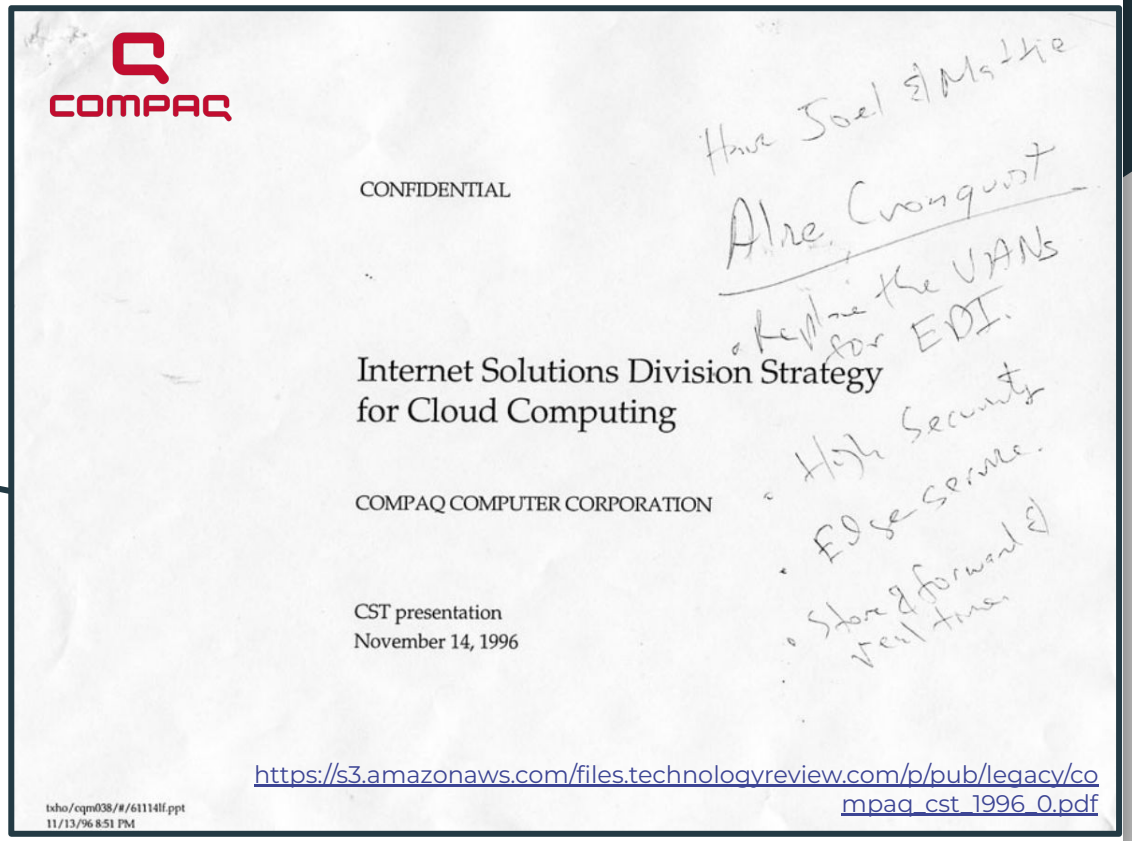
https://www.google.com/press/podium/ses2006.html

Who coined the term “Cloud Computing” ?

*George Favaloro holds Compaq’s 1996 Business Plan. This is the earliest printed document that mentions the term “**cloud computing**”.*



https://prog.world/who-coined-the-term-cloud-computing/



COMPAQ

CONFIDENTIAL

Internet Solutions Division Strategy
for Cloud Computing

COMPAQ COMPUTER CORPORATION

CST presentation
November 14, 1996

txho/cqm038/#/61114lf.ppt
11/13/96 8:51 PM

https://s3.amazonaws.com/files.technologyreview.com/p/pub/legacy/compaq_cst_1996_0.pdf

Cloud - Definitions

Cloud computing is the **on-demand availability of computer system resources**, especially data storage (cloud storage) and computing power, without direct active management by the user.

https://en.wikipedia.org/wiki/Cloud_computing

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(4) **Cloud.** a network of **servers** (= computers that control or supply information to other computers) on which **data** and **software** can be stored or managed and to which **users have access over the internet**

https://www.oxfordlearnersdictionaries.com/

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Le ***cloud computing*** (en français, « **informatique dans les nuages** ») fait référence à l'utilisation de la mémoire et des capacités de calcul **des ordinateurs et des serveurs répartis dans le monde entier et liés par un réseau**.

https://www.cnil.fr/fr/definition/cloud-computing

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https://www.oxfordlearnersdictionaries.com/

Cloud computing is the increasingly common practice of providing **'on-demand'** cloud services as IT infrastructure and resources.

https://www.ovhcloud.com/en/public-cloud/cloud-computing/

Cloud - Definitions

Cloud computing is a model for enabling **ubiquitous**, **convenient**, **on-demand** network access to a **shared pool of configurable computing resources** (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with **minimal management effort** or service provider interaction.

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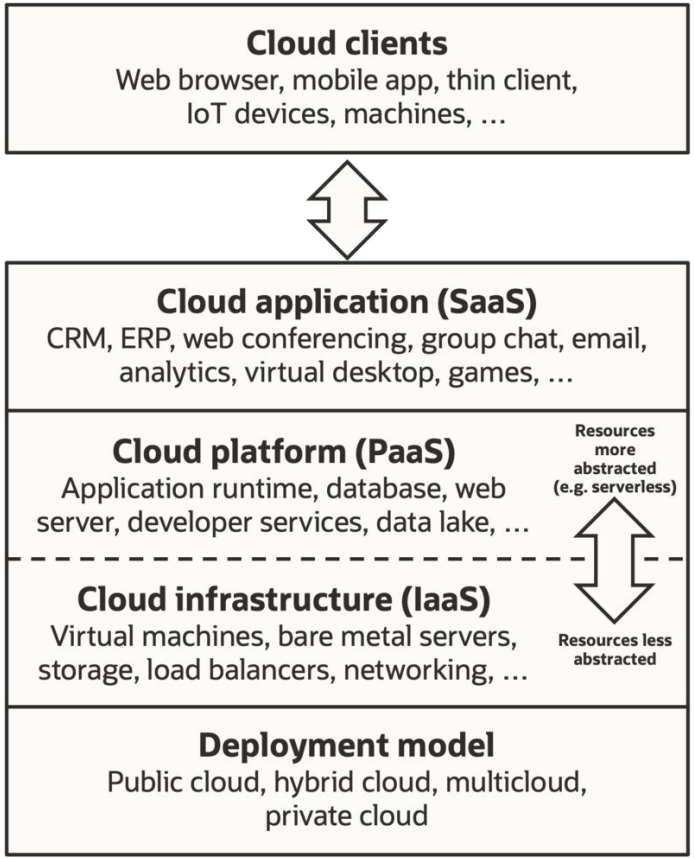
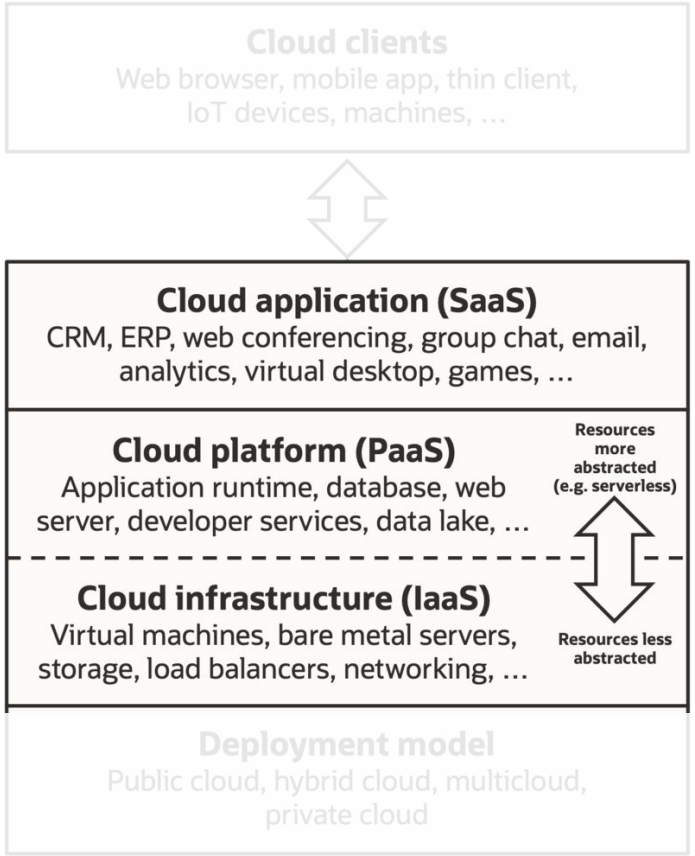
https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-145.pdf
(NIST - National Institute of Standards and Technology - 2011)

Cloud - Service Models



https://en.wikipedia.org/wiki/Cloud_computing

Cloud - Service Models



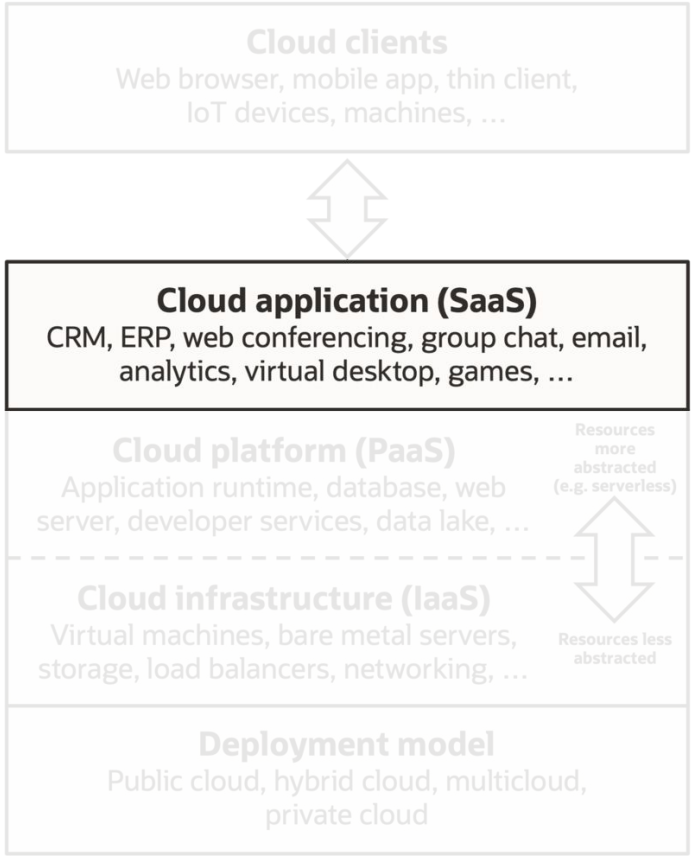
https://en.wikipedia.org/wiki/Cloud_computing

Cloud - Provided Services

Software as as Service (SaaS)

The capability provided to the consumer is to use the provider's applications running on a cloud infrastructure . The applications are accessible from various client devices [...]. The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user specific application configuration settings.

Examples: **Gmail, Slack, Google Drive, etc.**



https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-145.pdf

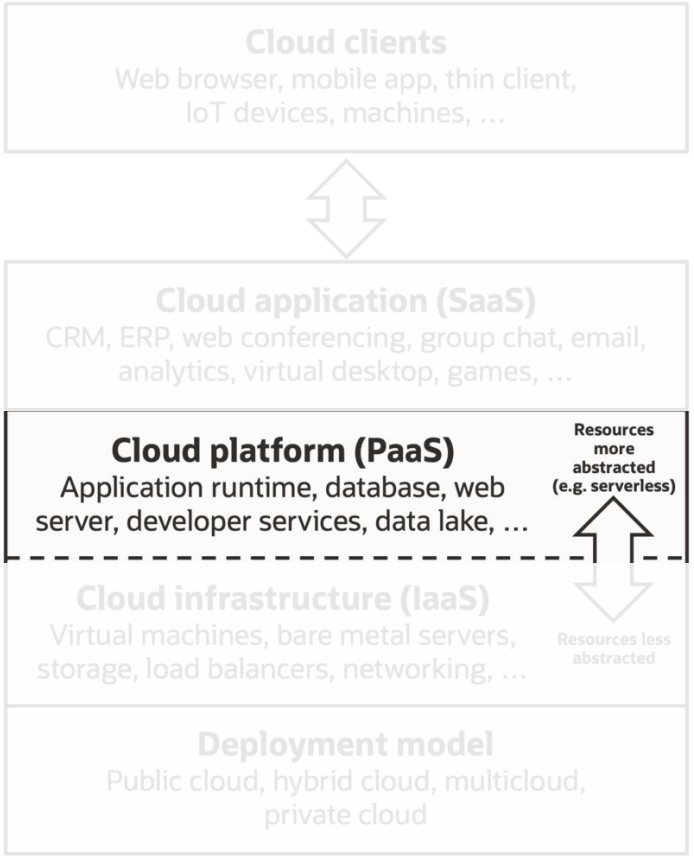
https://en.wikipedia.org/wiki/Cloud_computing

Cloud - Provided Services

Platform as as Service (PaaS)

Platform as a Service (PaaS). The capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages, libraries, services, and tools supported by the provider. The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has control over the deployed applications and possibly configuration settings for the application-hosting environment.

Examples: Heroku, Google App Engine, Amazon Lambda (FaaS - Function as a Service), …



https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-145.pdf

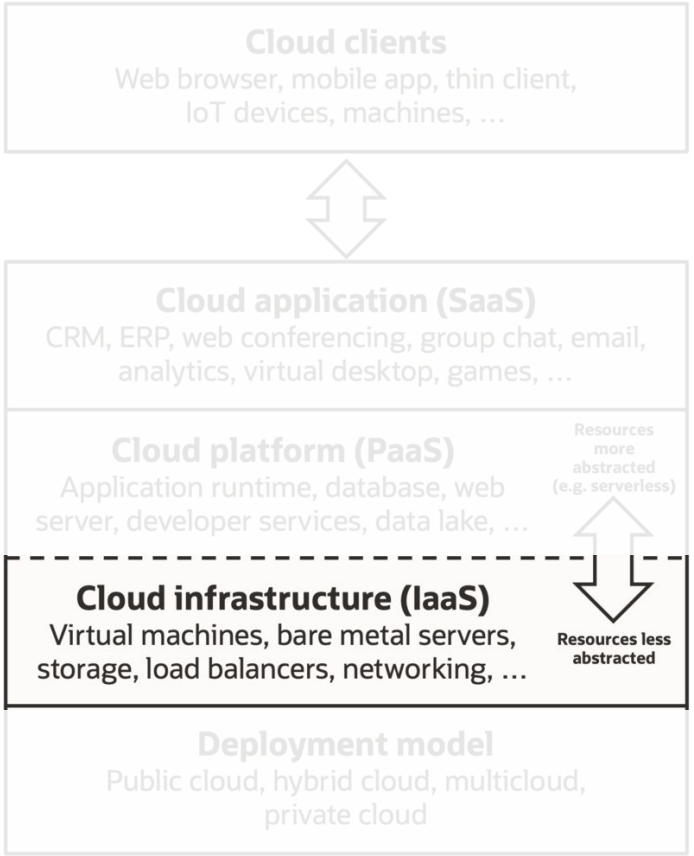
https://en.wikipedia.org/wiki/Cloud_computing

Cloud - Provided Services

Infrastructure as as Service (IaaS)

The capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications. The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, and deployed applications; and possibly limited control of select networking components (e.g., host firewalls).

Examples: Google Cloud Platform (GCP), Amazon Web Services (AWS), Google Azure, OVH Public Cloud, ...



https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-145.pdf

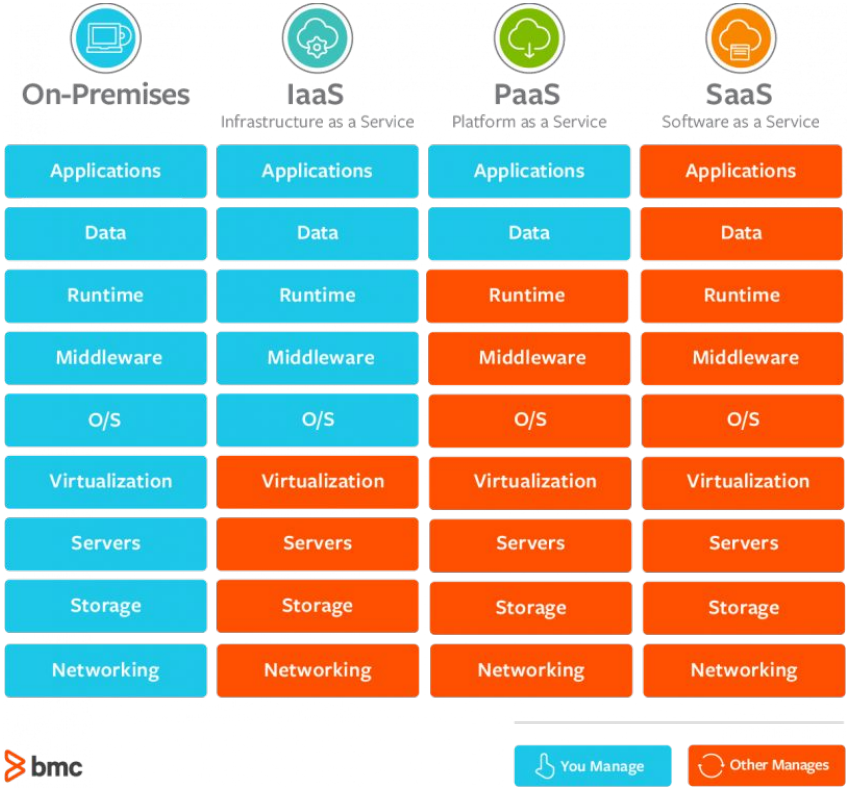
https://en.wikipedia.org/wiki/Cloud_computing

Cloud - Cloud Computing vs On-Premise

On-Premise

On-premises refers to IT infrastructure hardware and software applications that are **hosted on-site**. This contrasts with IT assets that are hosted by a public cloud platform or remote data center.

https://www.insight.com/en_US/content-and-resources/glossary/o/on-premises.html



https://www.bmc.com/blogs/saas-vs-paas-vs-iaas-whats-the-difference-and-how-to-choose/

Cloud - Deployment Models

Public clouds

A cloud environment created from resources not owned by the end user that can be redistributed to other tenants.

Private clouds

Loosely defined as a cloud environment solely dedicated to the end user, usually within the user's firewall and sometimes on premise.

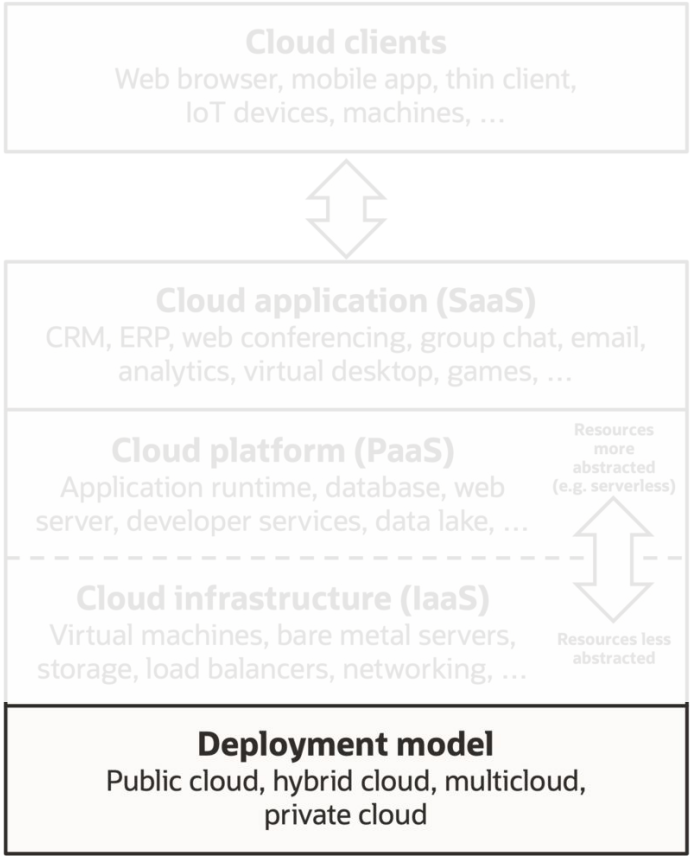
Hybrid clouds

Multiple cloud environments with some degree of workload portability, orchestration, and management among them.

Multiclouds

An IT system that includes more than 1 cloud—public or private—that may or may not be networked together.

https://www.redhat.com/en/topics/cloud



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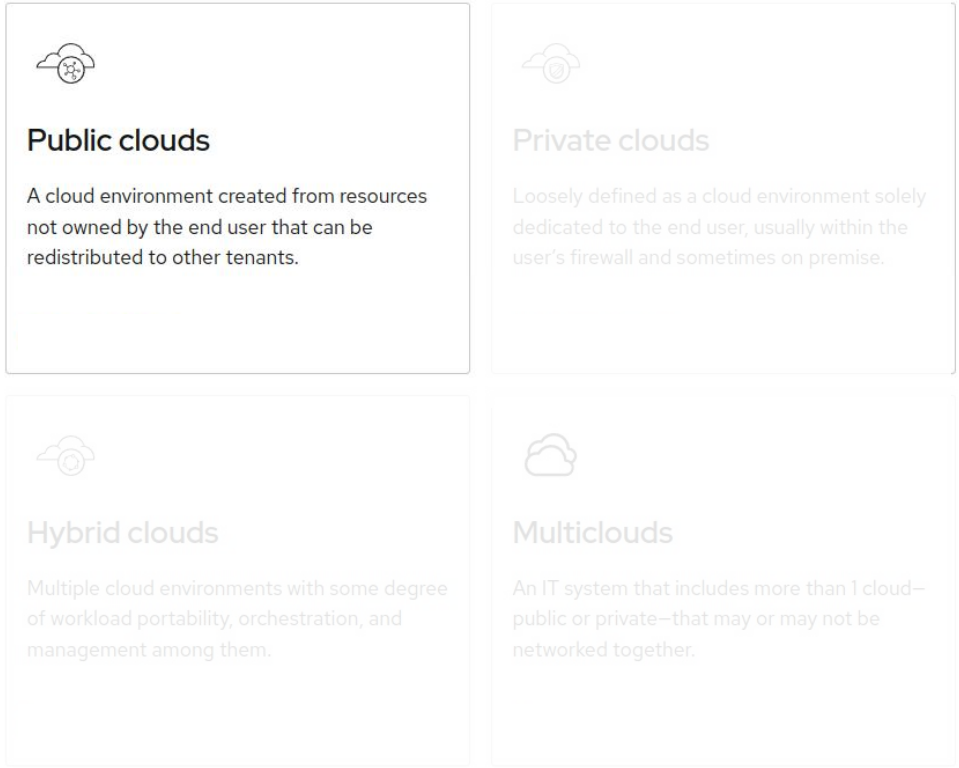
Cloud - Deployment Models

Public Clouds

A public cloud is a pool of virtual resources—developed from hardware owned and managed by a third-party company—that is automatically provisioned and allocated among multiple clients through a self-service interface.

Examples: Google Cloud Platform (GCP), Amazon Web Services (AWS), Google Azure, OVH Public Cloud, ...



https://www.redhat.com/en/topics/cloud

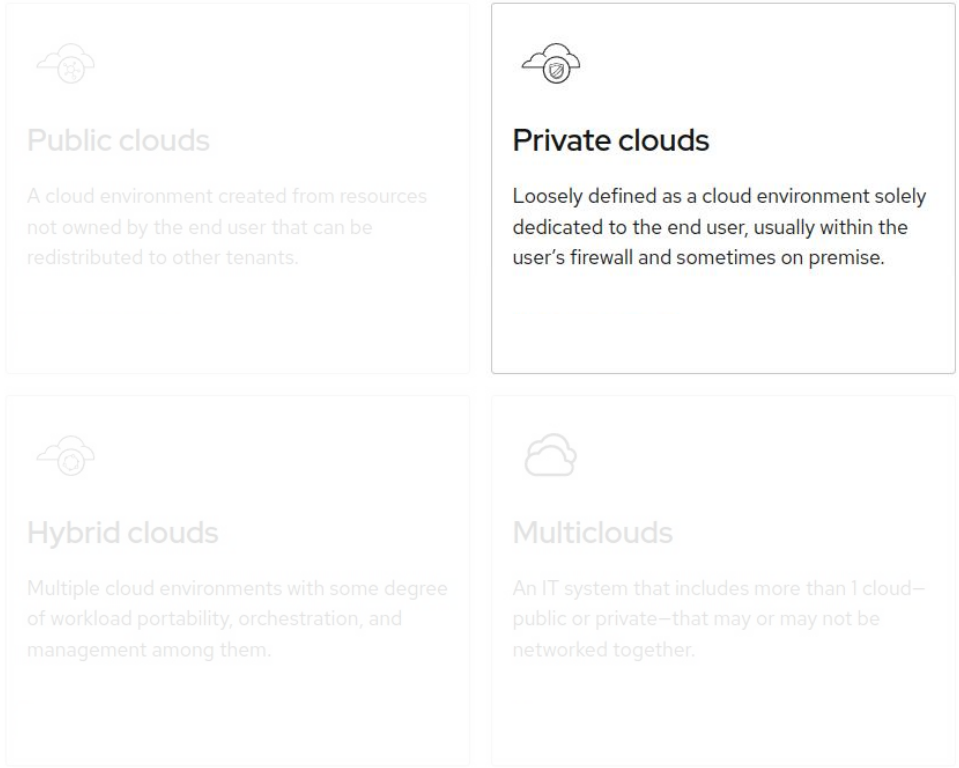
Cloud - Deployment Models

Private Clouds

Private Cloud are loosely defined as cloud environments solely dedicated to a single end user or group.

Private clouds no longer have to be sourced from **on-premise IT infrastructure**. Organizations are now building private clouds on rented, vendor-owned data centers located off-premises. This has also led to a number of private cloud subtypes, including **managed private clouds** and **dedicated clouds.**

Examples: OVH Hosted Private Cloud.

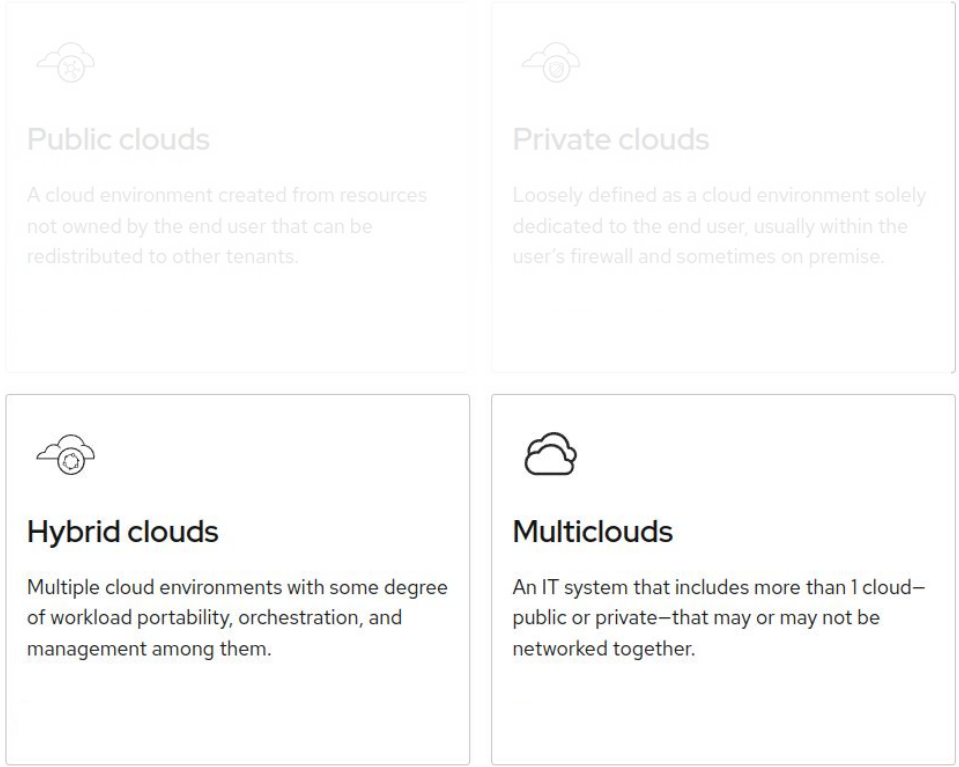


https://www.redhat.com/en/topics/cloud

Cloud - Deployment Models

Multiclouds and Hybrid Clouds

Multiclouds are a cloud approach made up of more than 1 cloud service, from **more than 1 cloud vendor** - public or private. All hybrid clouds are multiclouds, but not all multiclouds are hybrid clouds. Multiclouds become **hybrid clouds** when **multiple clouds are connected by some form of integration or orchestration**.



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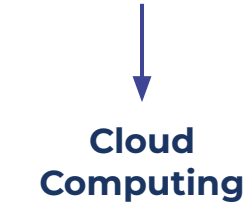
Section 3

Cloud and Data Engineering

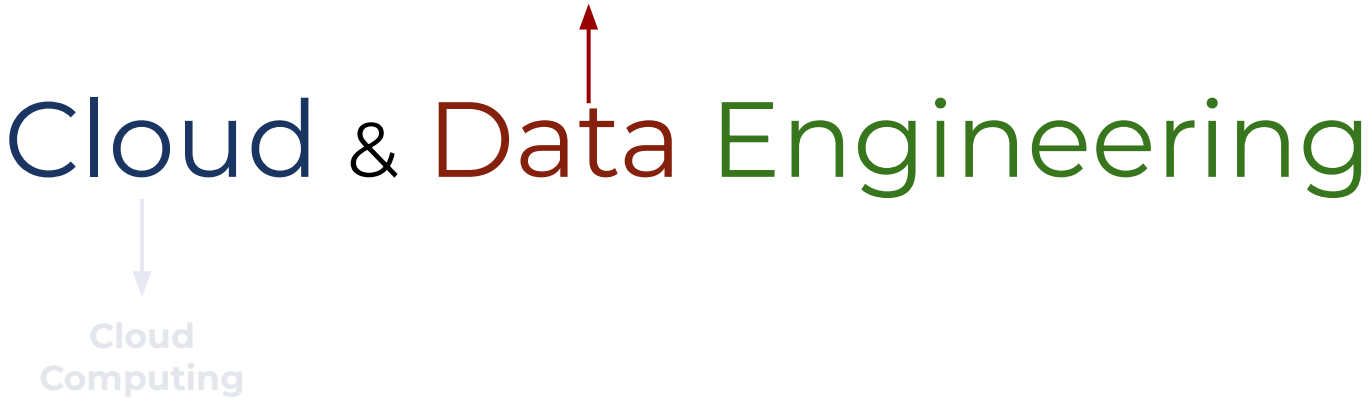
Cloud & Data Engineering

Cloud & Data Engineering



Cloud & Data Engineering

Data is the physical representation of information in a manner suitable for communication, interpretation, or processing by human beings or by **automatic means**.



Cloud & Data Engineering

Data is the physical representation of information in a manner suitable for communication, interpretation, or processing by human beings or by **automatic means**.

Cloud & Data Engineering

Cloud Computing

* The **creative application of scientific principles** to **design** or **develop structures**, machines, apparatus, or manufacturing processes, or works utilizing them singly or in combination; or to construct or operate the same with full cognizance of their design; or to forecast their behavior under specific operating conditions; all as respects an intended function, economics of operation and safety to life and property.

https://en.wikipedia.org/wiki/Engineering#cite_note-ABET_History-3

Cloud & Data Engineering

Cloud Engineering

Cloud engineering is the application of **engineering disciplines** to **cloud computing**. It brings a systematic approach to concerns of commercialization, standardization, and governance of cloud computing applications. In practice, it leverages the methods and tools of engineering in **conceiving**, **developing**, **operating** and **maintaining cloud computing systems** and **solutions**.

https://en.wikipedia.org/wiki/Cloud_engineering

Data Engineering

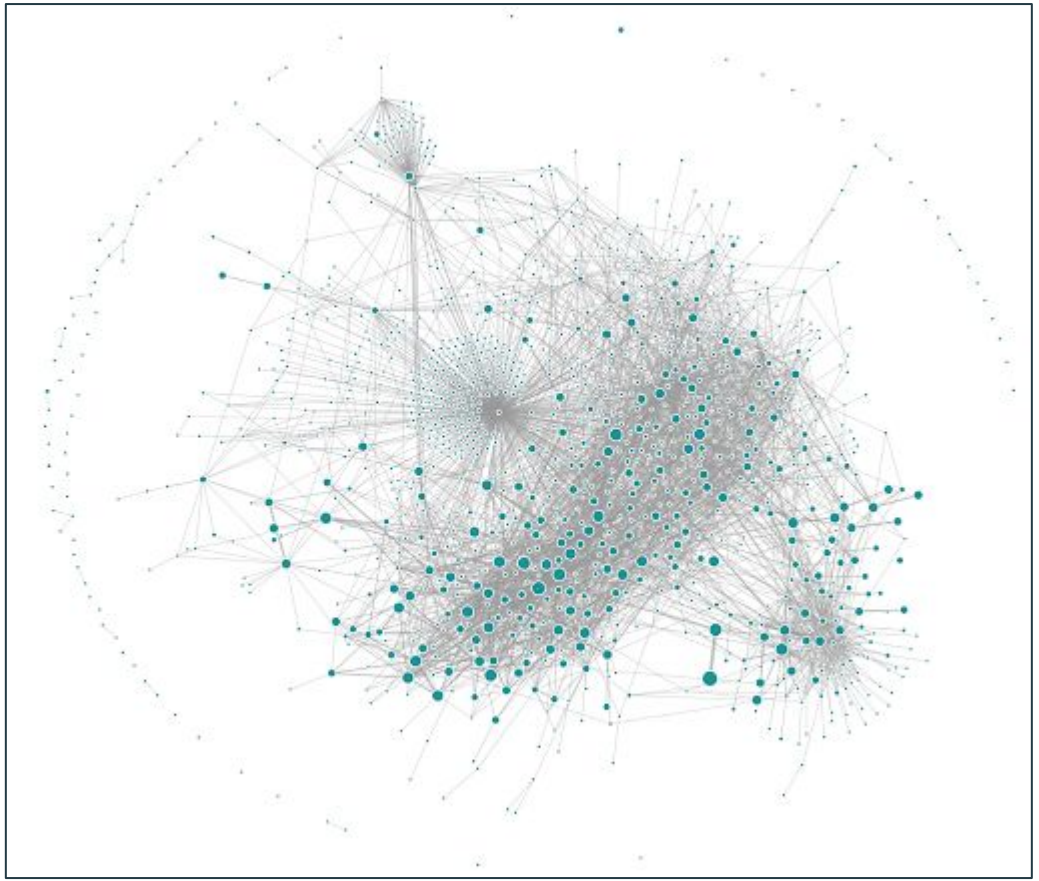
Data engineering refers to the building of systems to enable the **collection** and **usage** of **data**. This data is usually used to enable subsequent analysis and data science; which often involves machine learning. Making the data usable usually involves substantial **compute** and **storage**, as well as **data processing**.

https://en.wikipedia.org/wiki/Data_engineering

Section 4

Cloud and Architecture - Part I

Uber's microservice architecture circa **mid-2018** from **Jaeger***

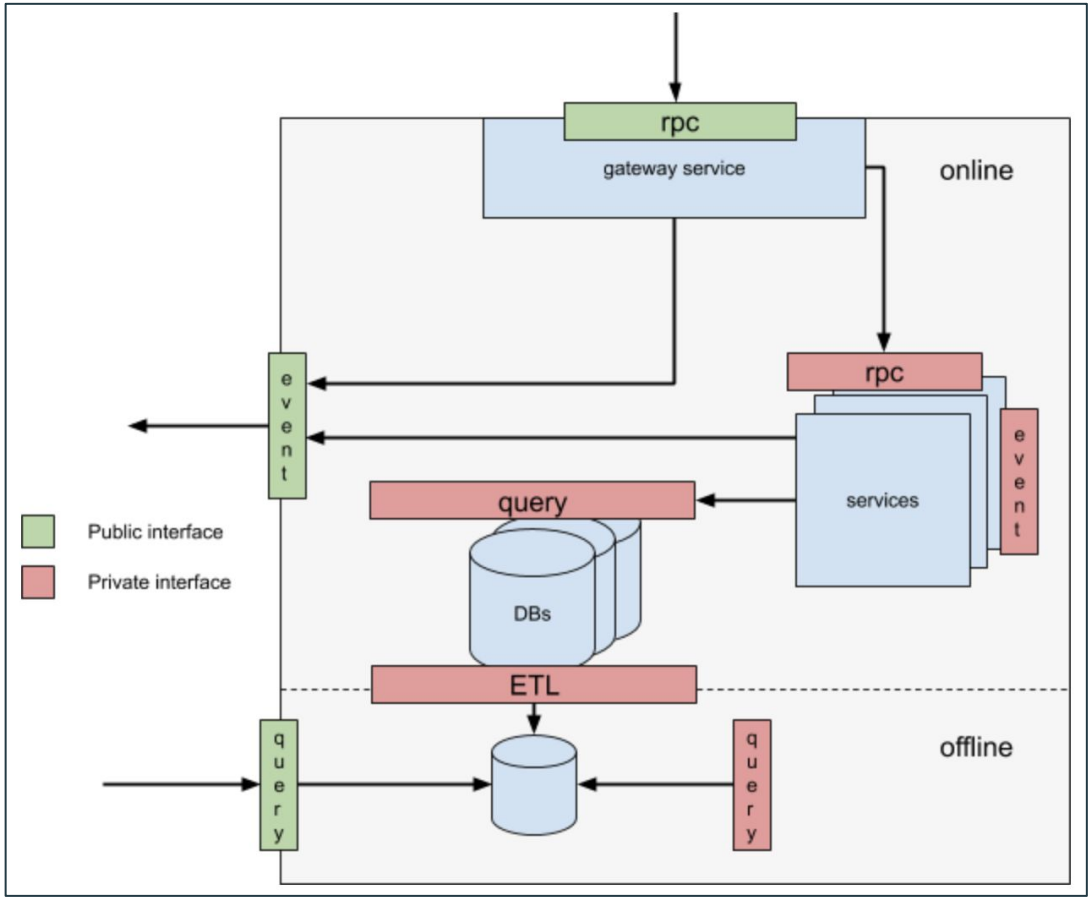


Uber

Jaeger** is open source software for tracing transactions between distributed **services

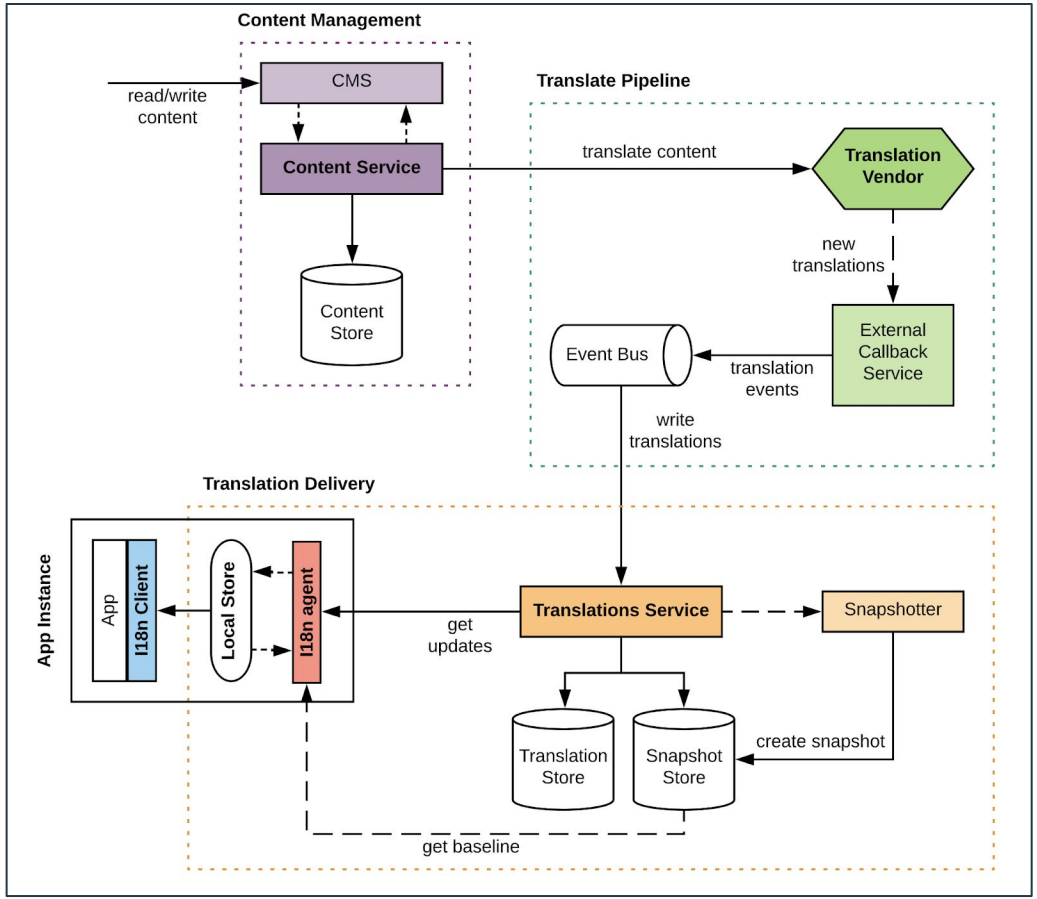
https://www.uber.com/en-FR/blog/microservice-architecture **[2020]**

Gateway API in a Domain-Oriented Microservice Architecture (Uber)



https://www.uber.com/en-FR/blog/microservice-architecture **[2020]**

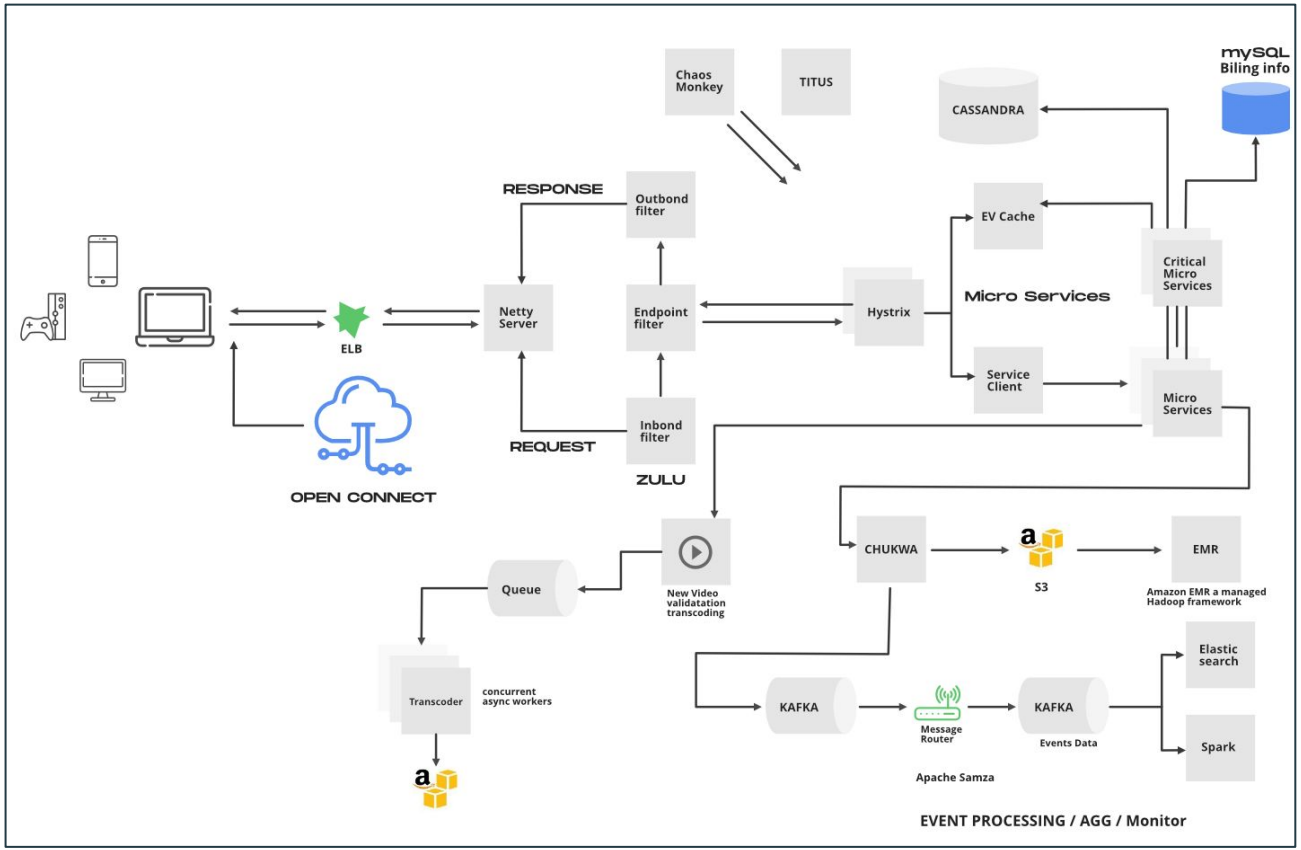
Airbnb's Internalization* Platform



***Internationalization** is the process of adapting software to accommodate for different languages, cultures, and regions (i.e. locales), while minimizing additional engineering changes required for localization.*

https://medium.com/airbnb-engineering/building-airbnbs-internationalization-platform-45cf0104b63c **[2020]**

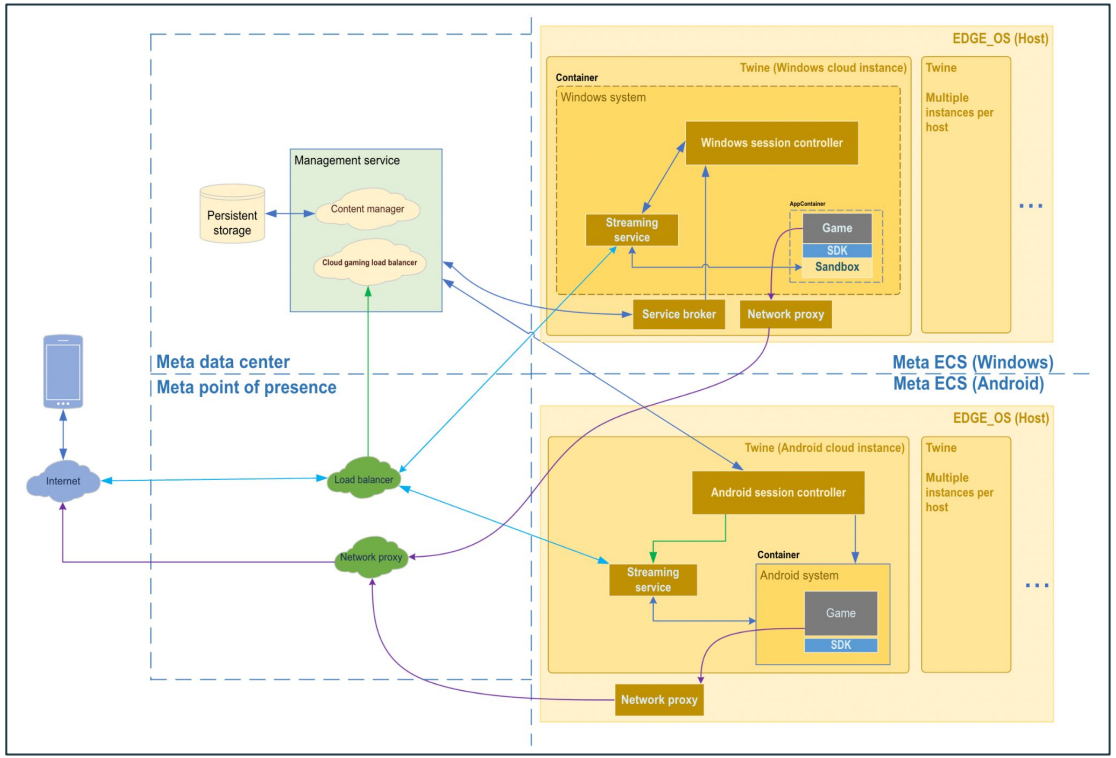
Netflix Backend Architecture



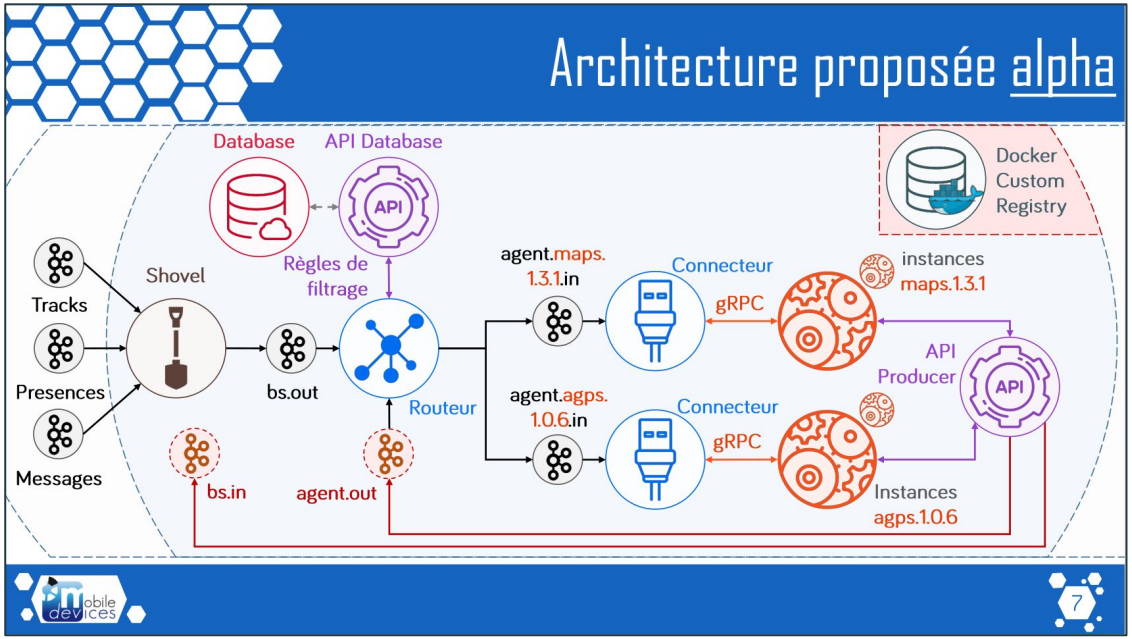
NETFLIX

https://dev.to/gbengelebs/netflix-system-design-backend-architecture-10i3 **[2020]**

Meta's Cloud Gaming Infrastructure



https://engineering.fb.com/2022/06/09/web/cloud-gaming-infrastructure/ **[2022]**



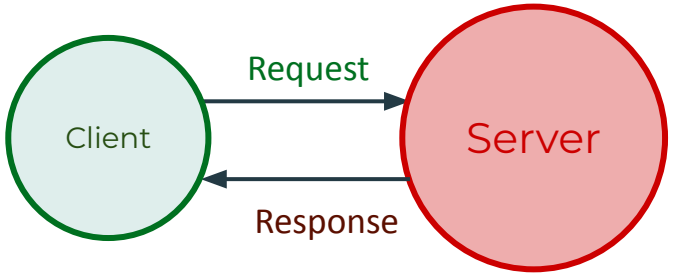
Munic - cloud V4 Draft (2016)

MUNIC
CAR DATA

Architecture Patterns - Building Patterns

Client-Server (request-response) Pattern

In the client-server architecture patterns, there are two main components: The client, which is the ***service requester***, and the server, which is the ***service provider***. Although both client and server may be located within the same system, they often communicate over a network on separate hardware.



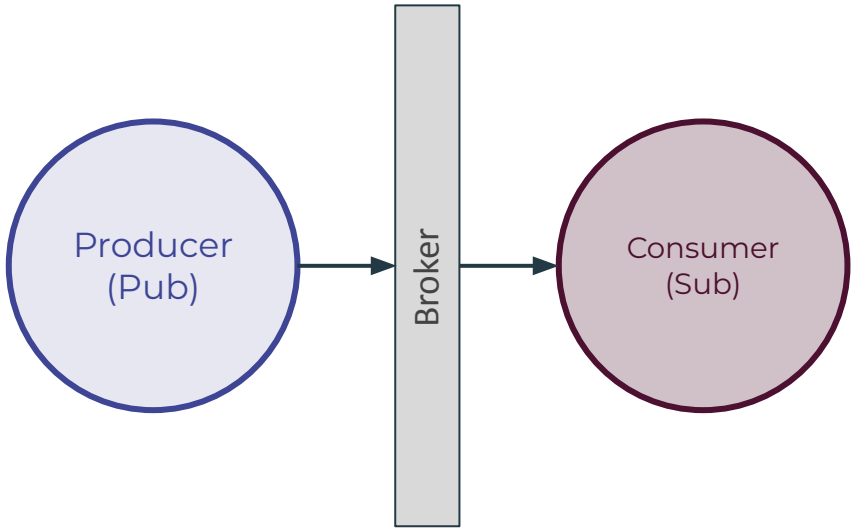
https://www.redhat.com/architect/5-essential-patterns-software-architecture

Architecture Patterns - Building Patterns

Publisher-Subscriber (Producer-Consumer) **Pattern**

Enable an application to **announce events** to multiple interested consumers **asynchronously***, **without coupling** the **senders (publishers)** to the **receivers (consumers)**.

*** Asynchronous communication** differs from **synchronous communication** in the way that in asynchronous communication the producer **does not need to wait** the receiver to handle the produced events/data.



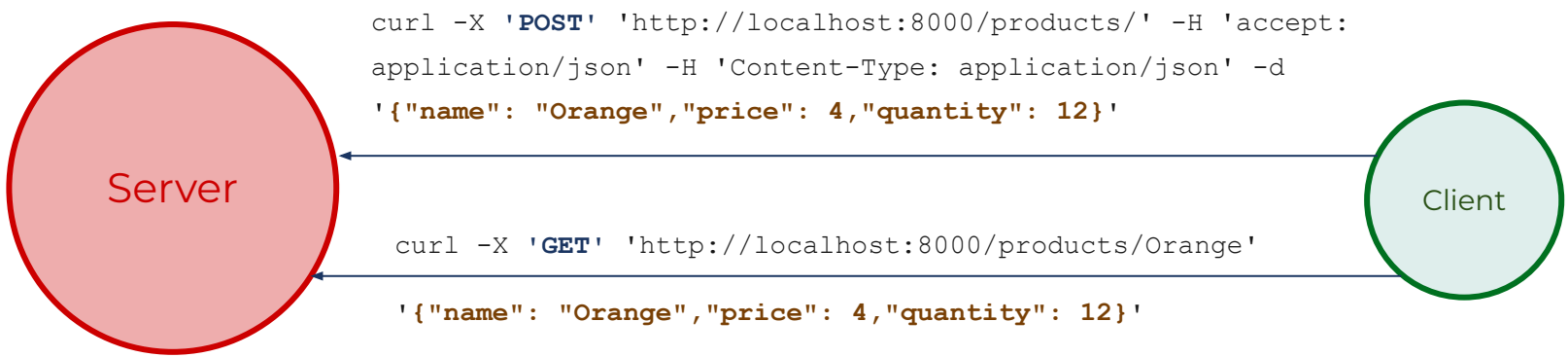
https://learn.microsoft.com/en-us/azure/architecture/patterns/publisher-subscriber

Section 5 - Exercice

Cloud and Architecture - Part II -
Implementing a Client-Server Pattern

Architecture Patterns - Client-Server Pattern

Exercice 1:
Implement a simple **REST API** to manage products in a store. A **Product** has a **name**, **price** and **quantities** for attributes. At Least **POST** and **GET** should be implemented.



Your REST API in Python or any other chosen language

cURL linux command line tool

Architecture Patterns - Client-Server Pattern

Exercice 1:

Implement a simple **REST API** to manage products in a store. A **Product** has a **name**, **price** and **quantities** for attributes. At Least **POST** and **GET** should be implemented.

Questions:

What is an API ? Why API and not just a Web Interface ? What a REST API ?

Technologies:

Use whatever language you want. If you are using Python, we suggest **fastapi** and **pydantic**.

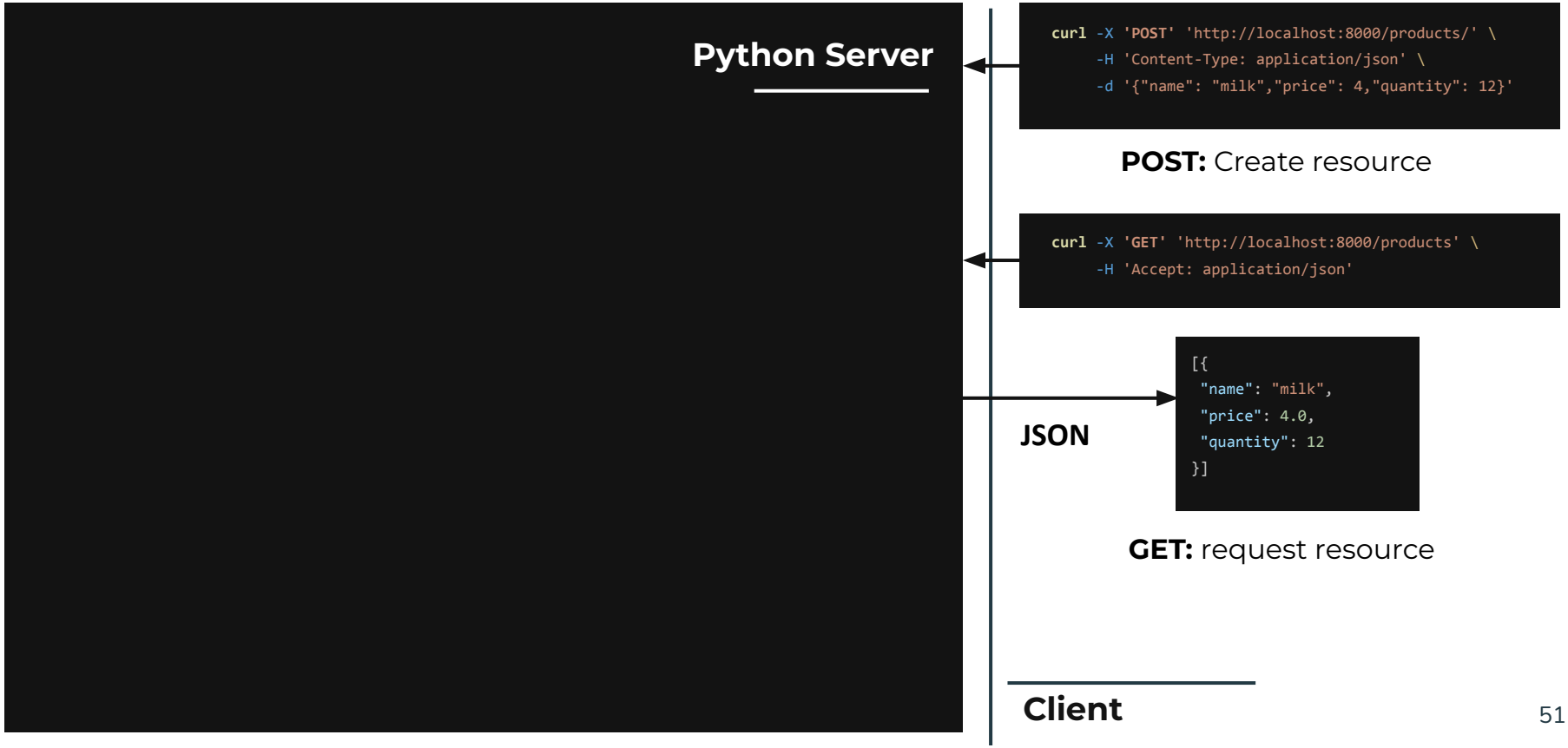
Warning:

For now, the data stored by your API does not need to be persistent.

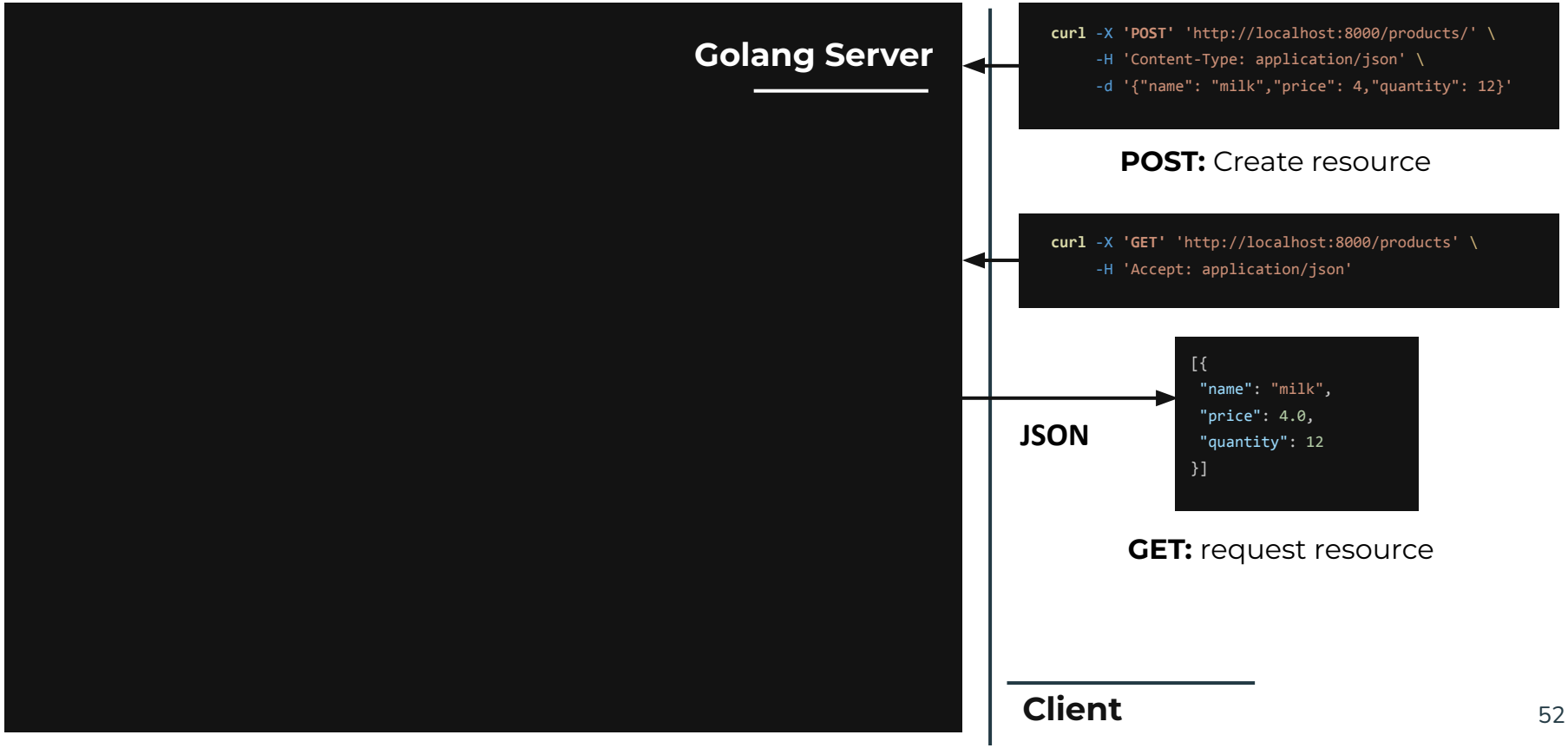
Section 5 - Solution

Cloud and Architecture - Part II -
Implementing a Client-Server Pattern

Architecture Patterns - Client-Server Pattern

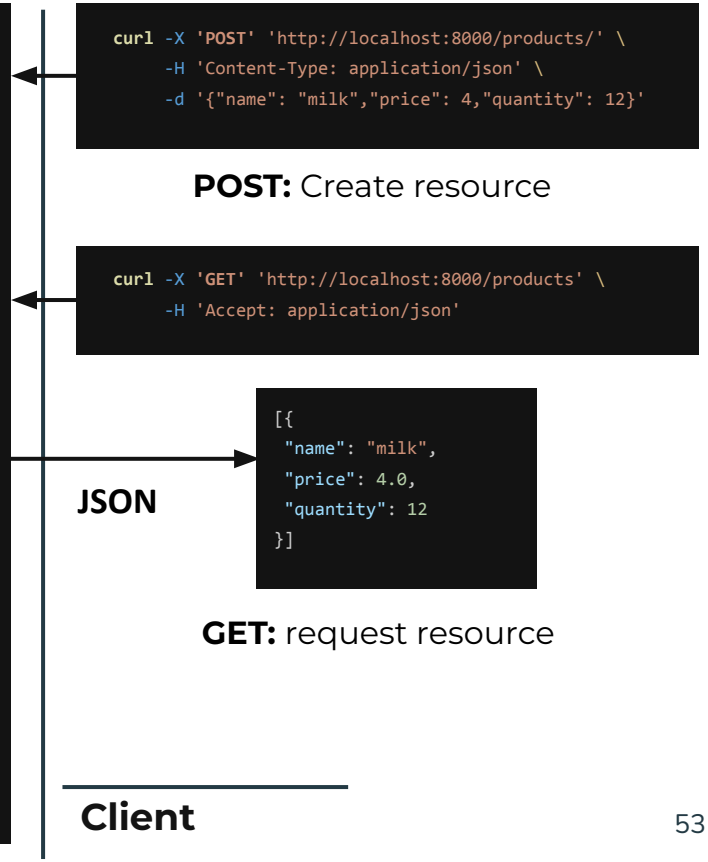


Architecture Patterns - Client-Server Pattern



Architecture Patterns - Client-Server Pattern

No matter what language is used



Architecture Patterns - Client-Server Pattern

No matter what language is used

What matters here:

Communication Protocol (set of rules):	Data Serialization Format:
HTTP	JSON

```
curl -X 'POST' 'http://localhost:8000/products/' \
     -H 'Content-Type: application/json' \
     -d '{"name": "milk","price": 4,"quantity": 12}'
```

POST: Create resource

```
curl -X 'GET' 'http://localhost:8000/products' \
     -H 'Accept: application/json'
```

JSON

```
[{
 "name": "milk",
 "price": 4.0,
 "quantity": 12
}]
```

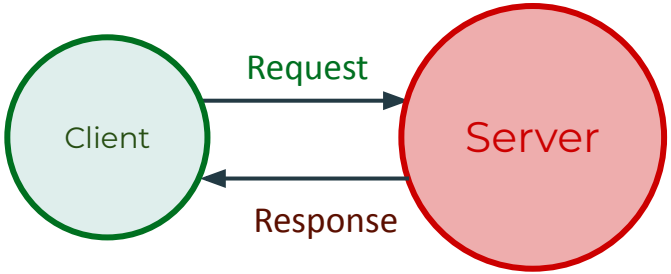
GET: request resource

Client

Architecture Patterns - Client-Server Pattern

API - Application Programming Interface:

API stands for **Application Programming Interface**, which is a set of definitions and protocols for building and integrating application software. **APIs** let your **product** or **service communicate** with **other products** and **services** without having to know how they're implemented.



https://www.redhat.com/en/topics/api/what-are-application-programming-interfaces

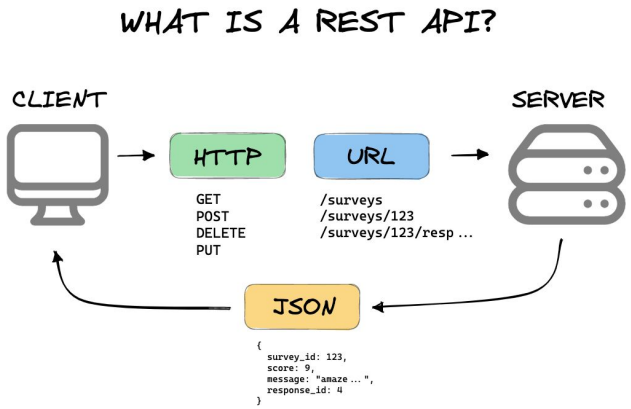
Architecture Patterns - Client-Server Pattern

RESTful API :

A **REST API** (a.k.a **RESTful API**) is an API that conforms to the constraints of **REST** architectural style. **REST** stands for **RE**presentational **S**tate **T**ransfer and is is a set of architectural constraints, not a protocol or a standard.

In order for an API to be considered **RESTful**, it has to conform to these criteria:

- Client-Server Architecture.
- Statelessness.
- Cacheability.
- Layered system.
- Uniform Interface.
- (Often) **HTTP** Based.



https://mannhowie.com/rest-api

https://ics.uci.edu/~fielding/pubs/dissertation/fielding_dissertation.pdf
https://www.redhat.com/en/topics/api/what-are-application-programming-interfaces

Architecture Patterns - Client-Server Pattern

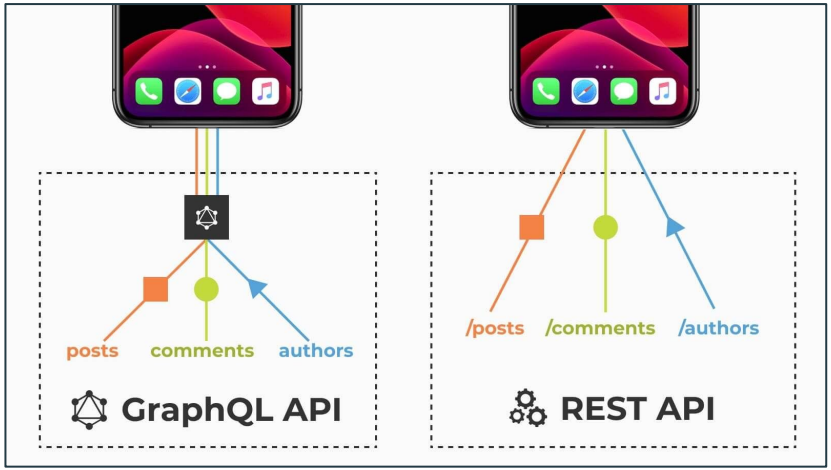
Note - REST API and GraphQL:

REST API is not the only way to implements APIs.

GraphQL is an open-source data query and manipulation language for APIs and a query runtime engine.

GraphQL enables declarative data fetching where a client can **specify exactly what data it needs from an API**. Instead of multiple endpoints that return separate data, a GraphQL server exposes a single endpoint and responds with precisely the data a client asked for.

https://en.wikipedia.org/wiki/GraphQL



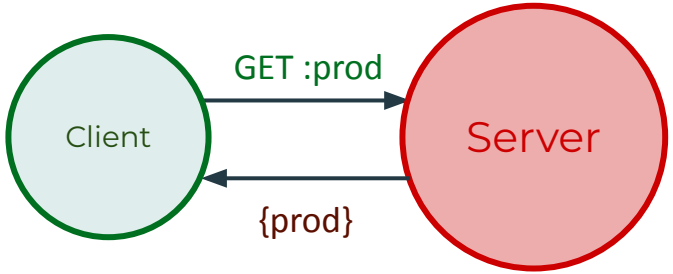
https://www.mobilelive.ca/blog/graphql-vs-rest-what-you-didnt-know

Section 7 - More components

Cloud and Architecture - Part III -
Adding a database in our Architecture

Persistence

Once our server is shutdown, all the products are **lost**. Our system is **not persistent !**



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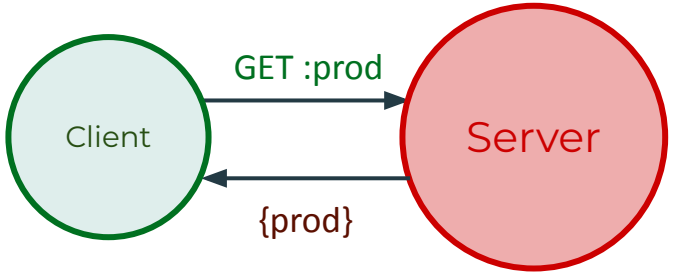

Persistence refers to the characteristic of state of a system that outlives (persists more than) the process that created it. This is achieved in practice by storing the state as data in **computer data storage**.

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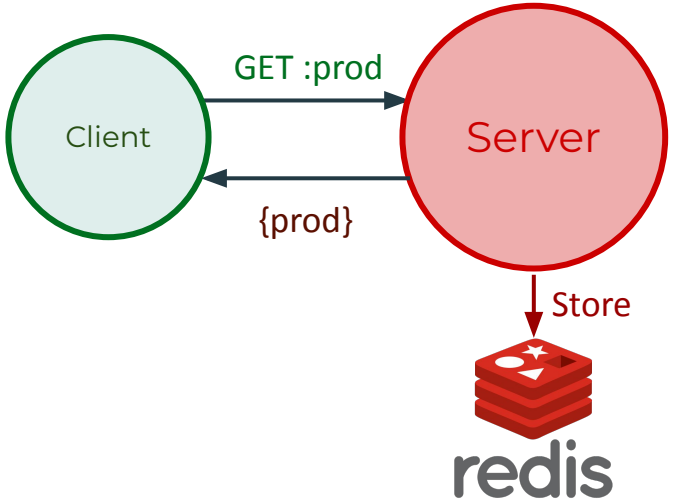
Exercice 2:

Make your system **persistent** using **redis key-value store (NoSQL Database)**.



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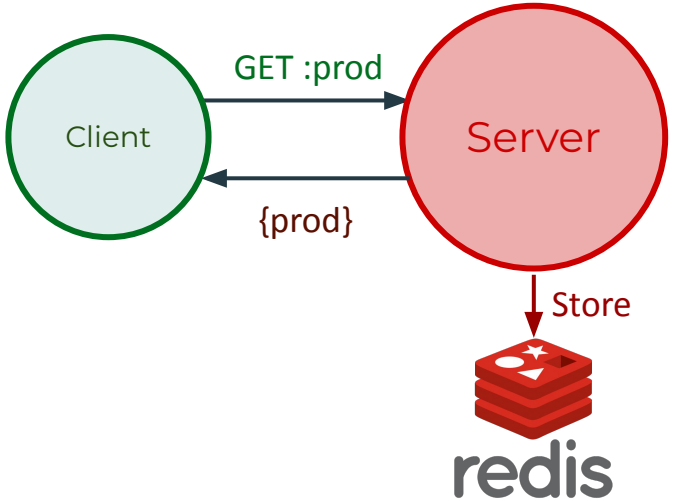

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Exercice 2:

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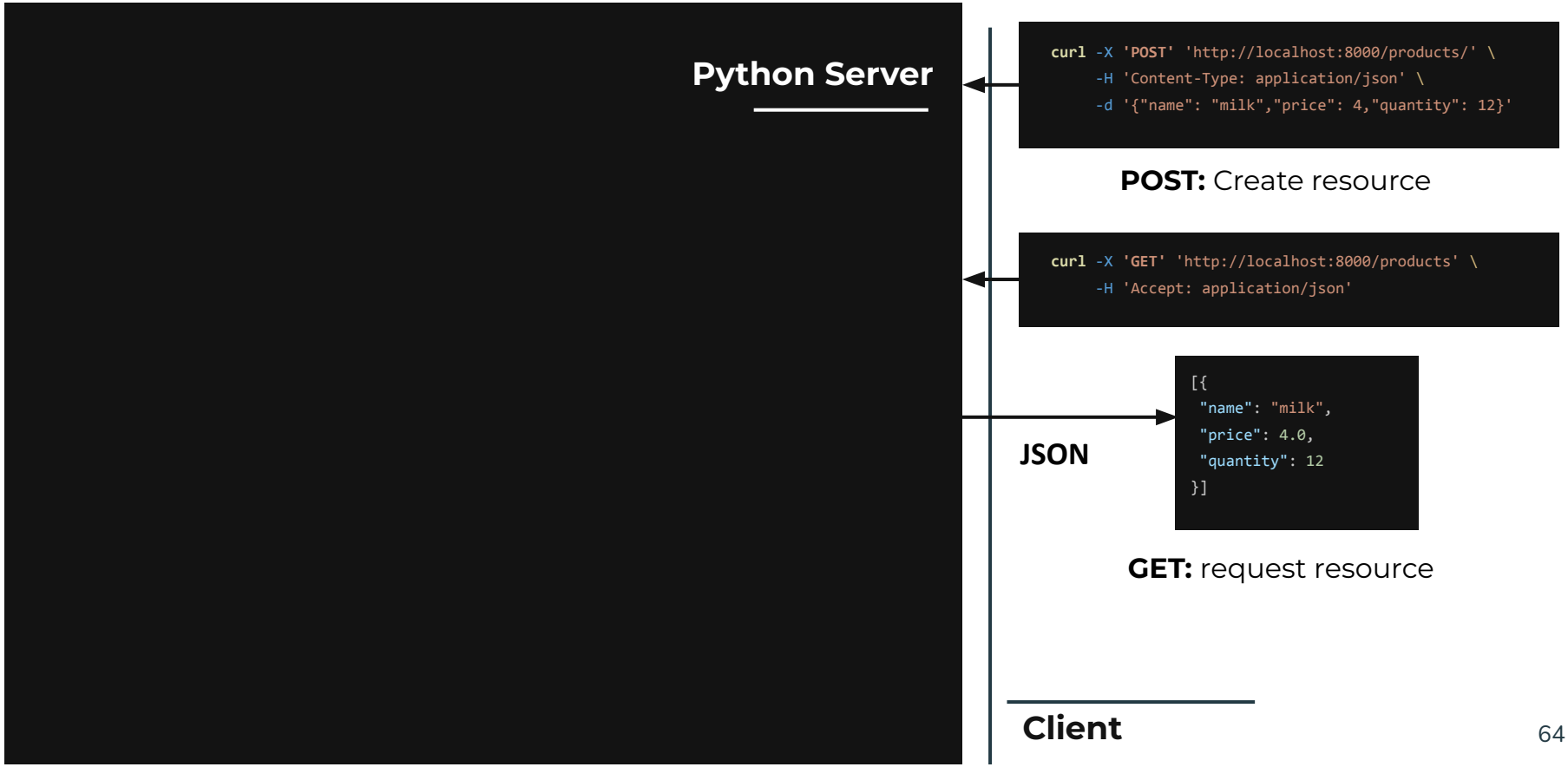
Questions:

- What is a NoSQL Database ?
- Enumerate and define at least 4 different kind of databases.
- What is a hosted database ?

Section 8 - Solution

Cloud and Architecture - Part III -
Adding a database in our Architecture

Architecture Patterns - Using Redis



NoSQL Databases

NoSQL, also referred to as “not only SQL”, “non-SQL”, is an approach to database design that enables the **storage and querying of data outside the traditional structures** found in relational databases.

*https://www.ibm.com/topics/nosql-databases

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Instead of the **typical tabular structure of a relational database**, NoSQL databases, house data within one data structure, such as JSON document. Since this non-relational database design **does not require a schema (less stringent constraints)**, it offers rapid scalability to manage large and typically **unstructured (semi-structured) data sets**.

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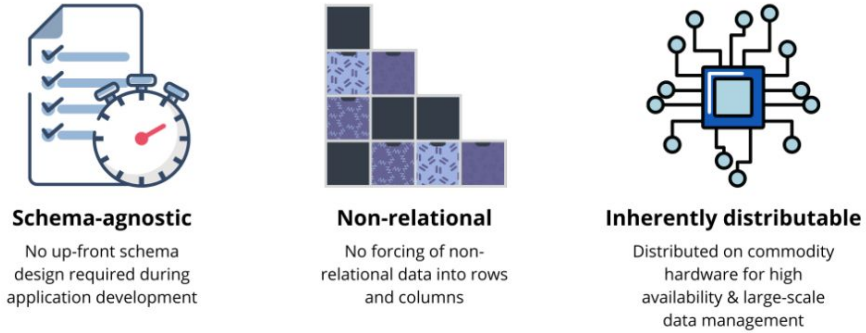

Main Characteristics of NoSQL Databases
*https://thedigitalskye.com/2021/03/09/7-must-know-ideas-about-nosql/

*https://www.ibm.com/topics/nosql-databases

NoSQL Databases - Examples

Key-value database

Deliver web advertisements
Store & retrieve user session data
High-speed in-memory data caching

Column-oriented database

Store parse data (i.e. columns don't have a value)
Record & analyze log files
Build data summaries

Document database

Publish digital content (Content Management System)
Manage unstructured data feeds
Join data streams

Graph database

Build a web of facts
Reconstruct processes
Manage social graphs (e.g. social networks, organized crime detection)

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NoSQL Databases - Examples

Key-value database

Deliver web advertisements
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High-speed in-memory data caching

E.g. Redis

Column-oriented database

Store parse data (i.e. columns don't have a value)
Record & analyze log files
Build data summaries

E.g. Cassandra

Document database

Publish digital content (Content Management System)
Manage unstructured data feeds
Join data streams

E.g. MongoDB

Graph database

Build a web of facts
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Manage social graphs (e.g. social networks, organized crime detection)

E.g. Neo4J

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Hosted Databases

In database hosting, **a third party** offers the **hardware** and **infrastructure** to run a database of the client's choosing, often in the cloud. They also configure the environment for **secure access**, **ensure resources are available to scale** the database as needed, and offer **managed services** based on requirements.

*https://www.mongodb.com/database-hosting

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Database as a service (DBaas): the provider maintains the system infrastructure and database and delivers it as a **fully managed cloud service**. The service covers high-level administrative functions, such as **database installation**, **configuration**, **maintenance** and **upgrades**. Additional tasks, such as **backups**, **patching**, **performance management** and **security**, typically are also handled by the provider.

*https://www.techtarget.com/searchdatamanagement/definition/database-as-a-service-DBaaS

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E.g. Oracle (Relational/SQL)

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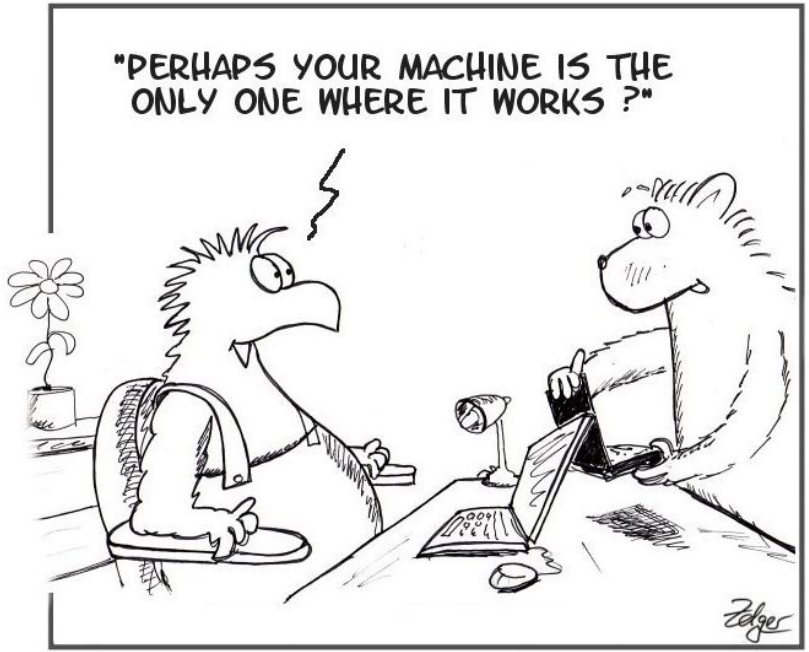
E.g. Oracle (Relational/SQL)

E.g. MongoDB (Document/NoSQL)

Section 9 - Portability

Well, try to run the code of your teammate !

Portability - Quick Overview



It works on my machine

*https://www.techtarget.com/searchstorage/definition/portability

Portability - Quick Overview

Porting is the task of doing any work necessary to make the computer program run in the new environment. Likewise, the term portability itself is a measure of whether software can be ported **without requiring major rework**.

*https://www.techtarget.com/searchstorage/definition/portability

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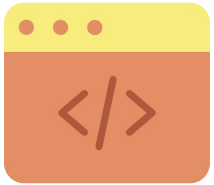
Porting is the task of doing any work necessary to make the computer program run in the new environment. Likewise, the term portability itself is a measure of whether software can be ported **without requiring major rework**.

For instance, a portable application could work on platforms such as Windows, Linux, iOS and Android. The porting process might include transferring installed program files to another system or creating the same program using different source code.

*https://www.techtarget.com/searchstorage/definition/portability

Portability - Types

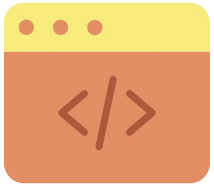


Source code portability

refers to building a program in a way that the same source code works in different environments. However, all the platforms must support the same programming language.

*https://www.techtarget.com/searchstorage/definition/portability

Portability - Types



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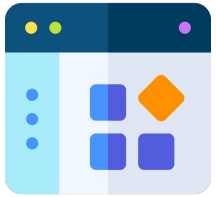

Application/SW portability

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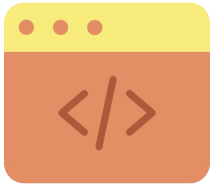

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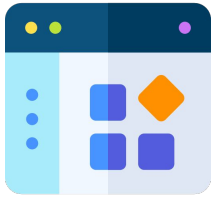

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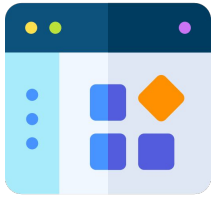

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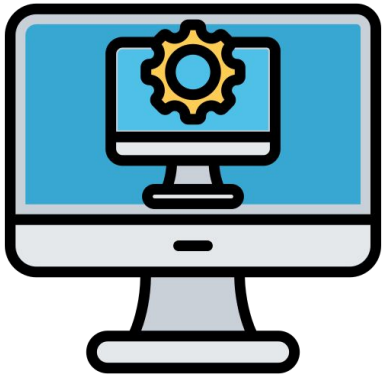
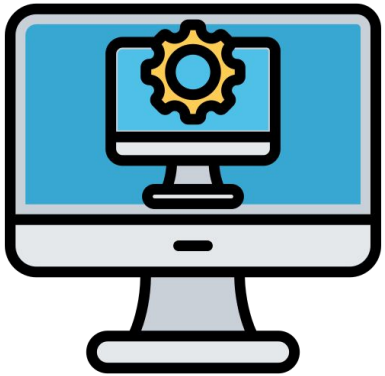
Virtualization



Virtualization is a process that allows for **more efficient utilization of physical computer hardware** and is the **foundation of cloud computing**.

https://www.ibm.com/topics/virtualization

Virtualization



Virtualization is a process that allows for **more efficient utilization of physical computer hardware** and is the **foundation of cloud computing**.

Virtualization uses software to create an **abstraction layer** over computer hardware that allows the hardware elements of a single computer—processors, memory, storage and more—to be **divided into multiple virtual computers**, commonly called virtual machines (VMs). Each VM runs **its own operating system (OS)** and ***behaves like an independent computer***, even though it is running on just a portion of the actual underlying computer hardware.

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Virtualization Benefits (1/2)

1. Resource Efficiency

Server virtualization lets you run several applications —each on its own VM with its own OS— on a single physical computer without sacrificing reliability. This enables maximum utilization of the physical hardware's computing capacity.

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2. Easier management

Replacing physical computers with software-defined VMs makes it easier to use and manage policies written in software. This allows you to create automated IT service management workflows. (e.g. automated deployment)

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Virtualization Benefits (2/2)

3. Minimal Downtime

OS and application crashes can cause downtime and disrupt user productivity. Admins can run multiple redundant virtual machines alongside each other and failover between them when problems arise. Running multiple redundant physical servers is more expensive.

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Virtualization Benefits (2/2)

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4. Faster Provisioning

Buying, installing, and configuring hardware for each application is time-consuming. Provided that the hardware is already in place, provisioning virtual machines to run all your applications is significantly faster (can be automated).

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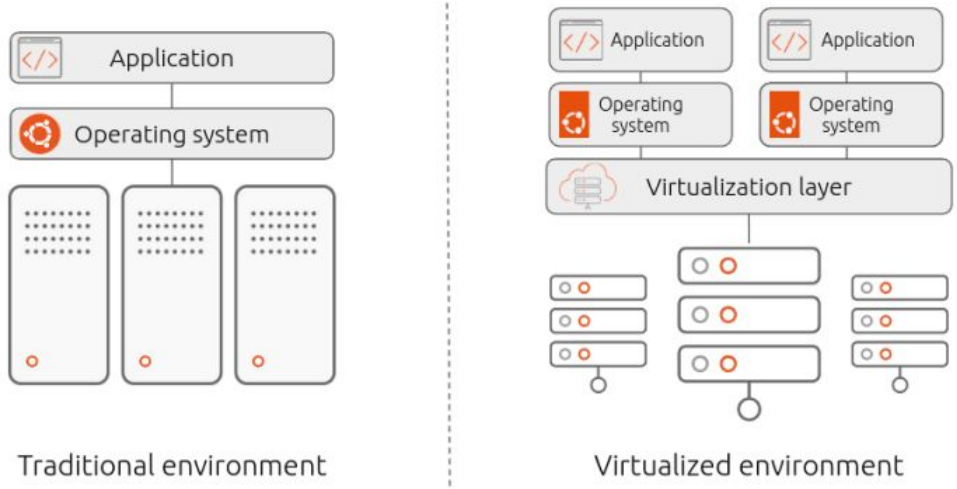
Types of Virtualization - in a nutshell

There are several ways of categorizing types of virtualization. The sheer amount of existing virtualization schemes and due to the nature of this course, we present here a short overview of existing types of virtualization.



https://ubuntu.com/blog/containerization-vs-virtualization

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1. **Hardware Virtualization :** The main job of hypervisor (in HW virtualization) is to control and monitor the processor, memory and other hardware resources.

2. **Operating System (OS) Virtualization :** In this kind of virtualization, the virtualization software is installed over the OS, and further use creates several other VMs (isolated user spaces).

3. **Data Virtualization :** Data virtualization tools create a software layer between the applications accessing the data and the systems storing it.

...Others: Network, Storage, Desktop, GPU, Application (e.g. JVM).

Hardware Virtualization

In a traditional physical computing environment, software such as an operating system (OS) or enterprise application has **direct access to the underlying computer hardware and components**, including the processor, memory, storage, certain chipsets and OS driver versions.

https://www.techtarget.com/searchitoperations/definition/hardware-virtualization

Hardware Virtualization

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Hardware virtualization installs a hypervisor or virtual machine manager (VMM), which creates an **abstraction layer between the software and the underlying hardware**. Once a **hypervisor** is in place, **software relies on virtual representations of the computing components**, such as virtual processors rather than physical processors. Popular hypervisors include VMware's vSphere, based on ESXi, and Microsoft's Hyper-V.

Type I Virtualization

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OS Virtualization

Unlike hardware-based virtualization, **OS virtualization is done over the top of the operating system**. This implies that the software installed on the OS virtualizes the same, making it the host.

https://blog.resellerclub.com/os-virtualization-vs-hardware-virtualization/#:~:text=While%20hardware%20virtualization%20wins%20for,KVM%20Hypervisor%2Dbased%20hardware%20virtualization.

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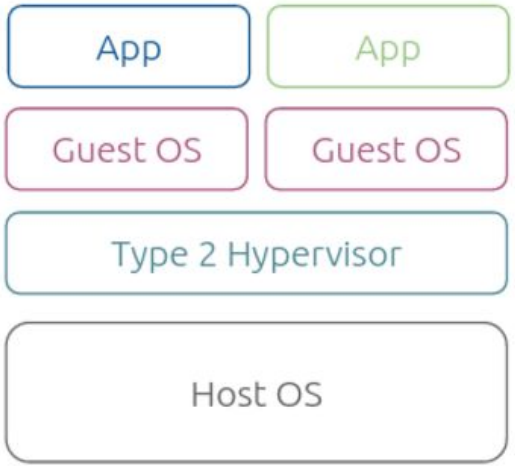
In this kind of virtualization, the virtualization software (VMM) is installed over the OS, and further **allows to the existence of multiple isolated VM.**. Such instances **may look like real computers from the point of view** of programs running in them. We distinguish:

1. **Type II Virtualization:** Less efficient than Type I Virtualization but more handy to use. We cite Vmware workstation and Oracle VirtualBox.



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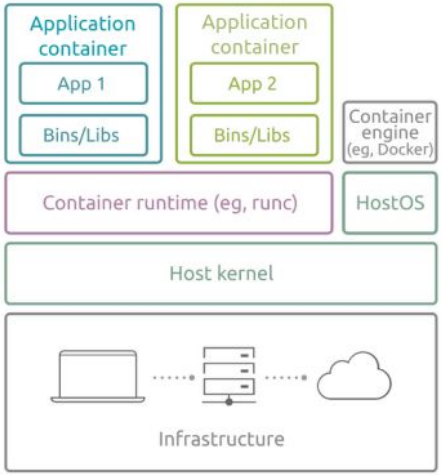
1. **Type II Virtualization:** Less efficient than Type I Virtualization but more handy to use. We cite Vmware workstation and Oracle VirtualBox.
2. **Containerization:** does not need an intermediate hypervisor and can rely directly on the host OS. We cite Docker (but there is a **jungle** of tools).



Application containers (eg, Docker)

Containerization

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Data Virtualization

Modern enterprises store data from **multiple applications**, using multiple file formats, in multiple locations, **ranging from the cloud to on-premise hardware and software systems**. Data virtualization lets any application access all of that data—**irrespective of source, format, or location**.

https://aws.amazon.com/what-is/virtualization/

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It is a method for **combining data** from various sources and different types **into a comprehensive, logical representation** without physically relocating the data. Simply put, users **can theoretically access and examine data while it still exists in its original sources thanks to dedicated middleware**. For instance Data Lake is a concept used to describe a form of such virtualization in contrast to Data Silos.

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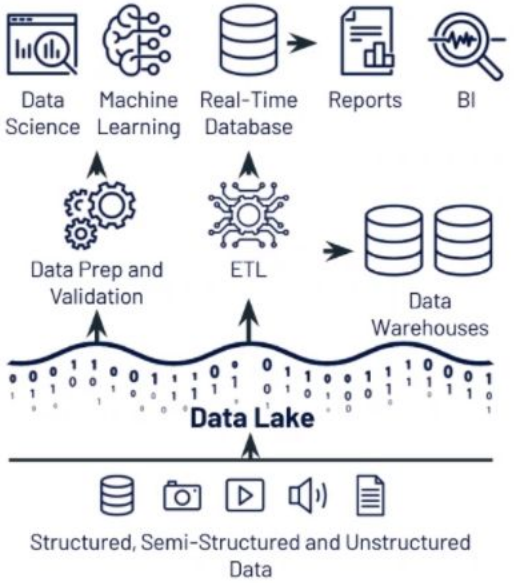

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Playing with Virtualization

Exercice 3:

Make your system portable (**Exercice 1**) using virtualization **(Use Docker)**

Questions:

What is Docker ?

Steps:

- Replace your installed redis by a **redis docker**.
- Write a **Dockerfile** to define your **docker image**.
- Run your docker image.
- Share your docker image with your teammate using hub.docker.com